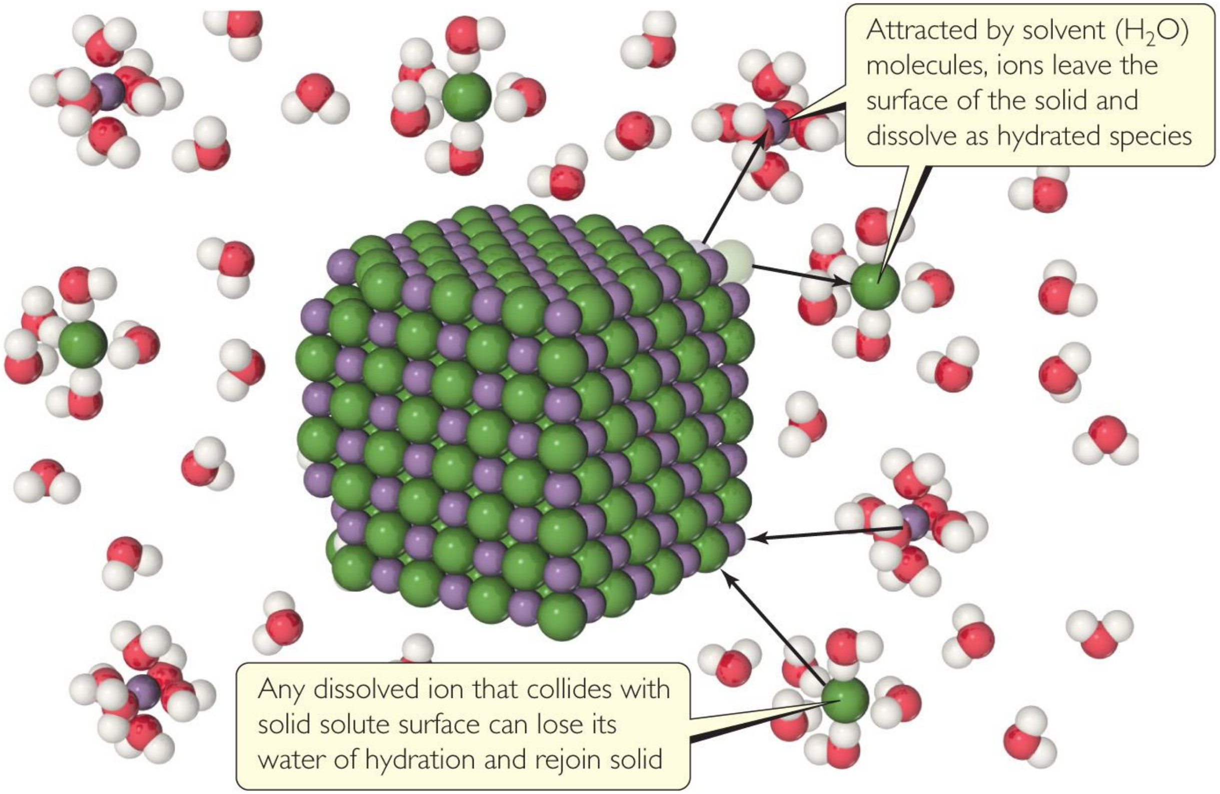
Chapter 13

Properties of Solutions

Solutions



- **Solutions (溶液)** are homogeneous mixtures of two or more pure substances. [sɒ'lju:t]
- In a solution, the **solute (溶质)** is dispersed (分散) uniformly throughout the **solvent (溶剂)**.
- **Solutions, however, can also be solids or gases.**

The solution process

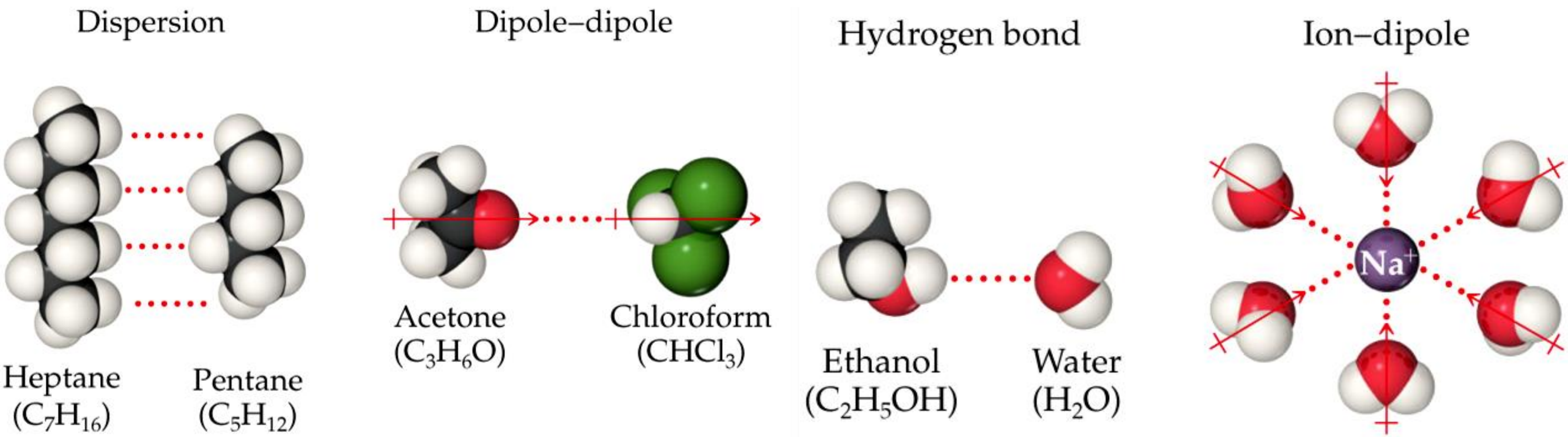
The ability of substances to form solutions depends on two factors:

(1) the natural tendency of substances to mix and spread into larger volumes when not restrained in some way

Entropy increase: (the *formation of solutions is favored by the increase in entropy that accompanies mixing*.)

(2) the types of intermolecular interactions involved in the solution process.

Intermolecular Interactions in Solutions



对于非极性物质，色散力占主导

对于离子物质，离子-偶极作用占主导

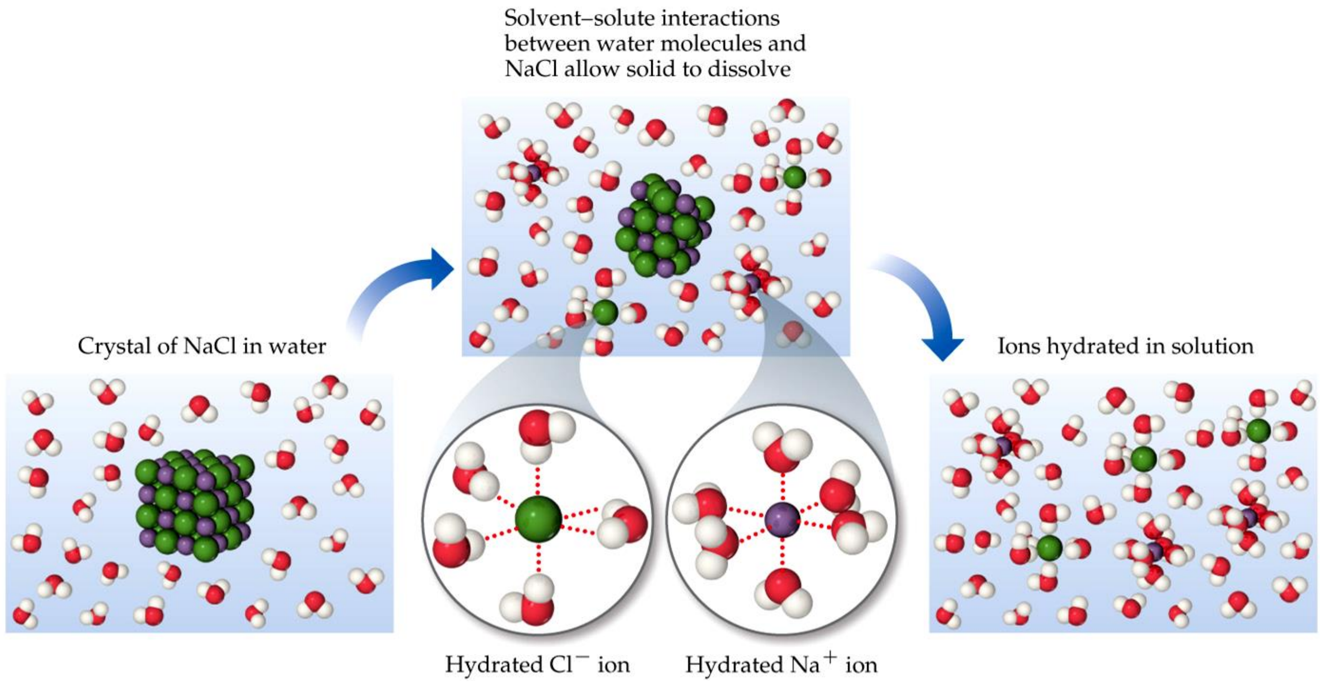
Dispersion forces dominate when one nonpolar substance, such as C_7H_{16}, dissolves in C_5H_{12}, and ion–dipole forces dominate in solutions of ionic substances in water.

How Does a Solution Form?

Three kinds of intermolecular interactions are involved in solution formation:

- **Solute–solute** interactions between solute particles must be overcome in order to disperse the solute particles through the solvent.
- **Solvent–solvent** interactions between solvent particles must be overcome to make room for the solute particles in the solvent.
- **Solvent–solute** interactions between solvent and solute particles occur as the particles mix.

Solutions

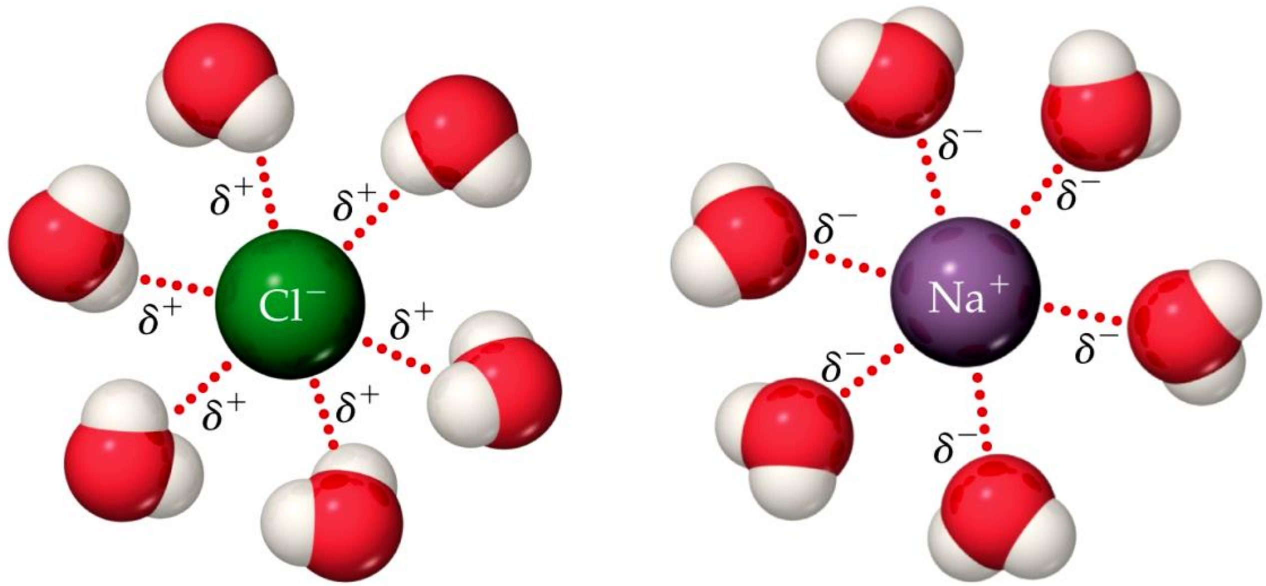


The intermolecular forces between solute and solvent particles must be strong enough to compete (竞争) with those between solute particles and those between solvent particles.

As a solution forms, the solvent pulls solute particles apart and surrounds-----called **solvation (溶剂化).**

If the solvent is water, the interactions are referred to as **hydration.**

How Does a Solution Form?



Positive ends of polar molecules are oriented toward negatively charged anion

Negative ends of polar molecules are oriented toward positively charged cation

If an ionic salt is soluble in water, it is because the ion–dipole interactions are strong enough to overcome the lattice energy of the salt crystal.

Lattice energy 晶格能又叫点阵能。它是在反应时1mol离子化合物中的阴、阳离子从相互分离的气态结合成离子晶体时所放出的能量。

Energy Changes in Solution

1. $(\text{solute})_n \longrightarrow \text{n solute}$ $\quad \Delta H_{solute}$

2. $(\text{solvent})_n \longrightarrow \text{n solvent}$ $\quad \Delta H_{solvent}$

3. $\text{n solute} + \text{n solvent} \longrightarrow \text{solution}$ $\quad \Delta H_{mix}$

---

4. $(\text{solute})_n + (\text{solvent})_n \longrightarrow \text{solution}$ $\quad \Delta H_{soln}$

$$\Delta H_{\text{soln}} = \underset{>0}{\Delta H_{\text{solute}}} + \underset{>0}{\Delta H_{\text{solvent}}} + \underset{<0}{\Delta H_{\text{mix}}}$$

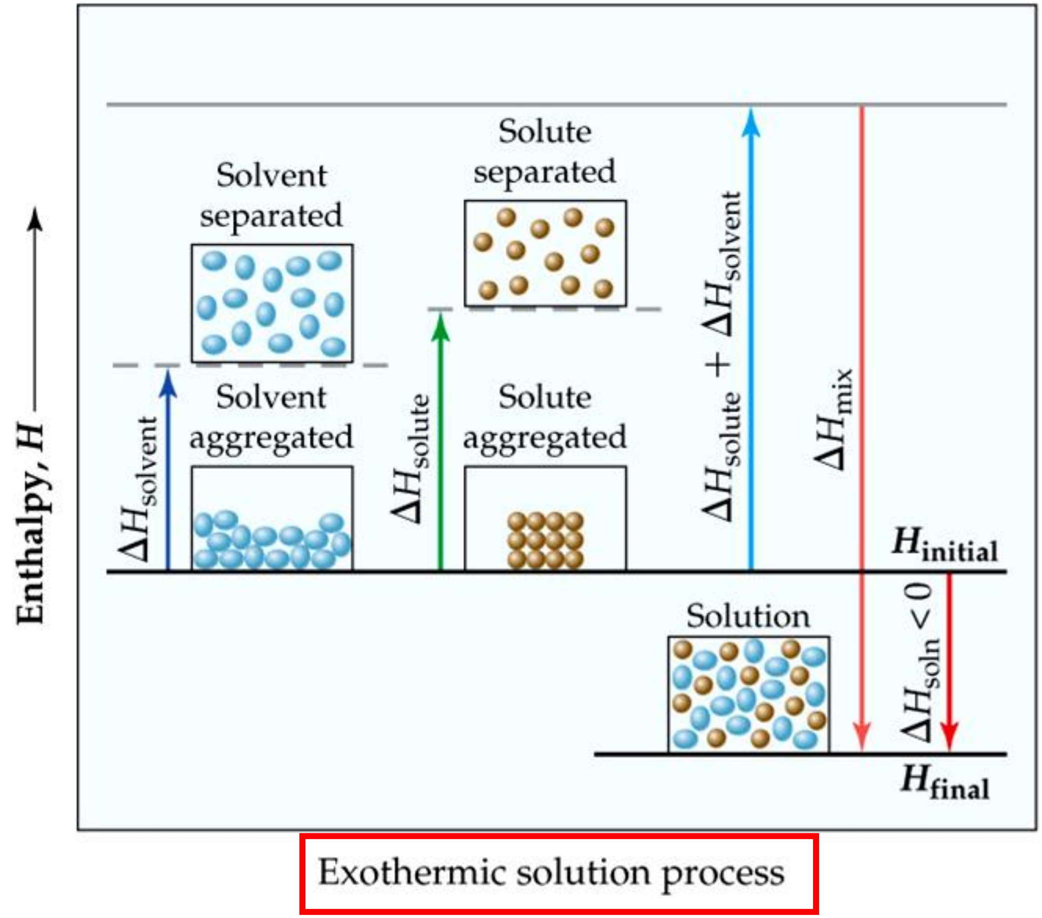
Energy Changes in Solution

[en'θælpɪ]

The enthalpy change of the overall process depends on ΔH for each of these steps.

ΔH_{solv} <0

Exothermic solution process



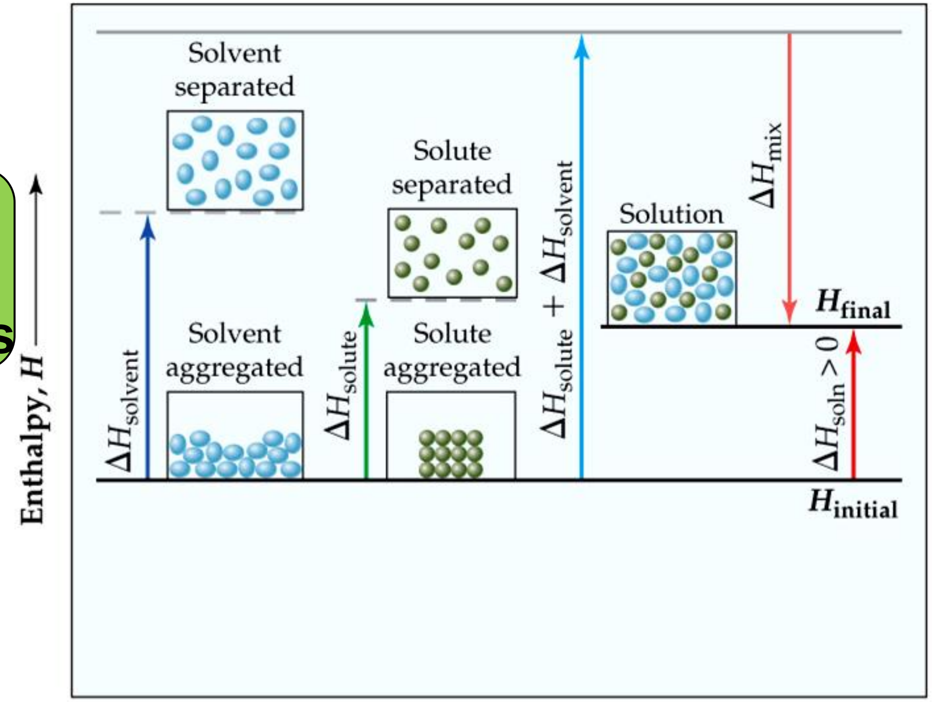
Magnesium Sulfate ($MgSO_4$)

Solutions

Why Do Endothermic Processes Occur?

$\Delta H_{solv} > 0$

Endothermic solution process

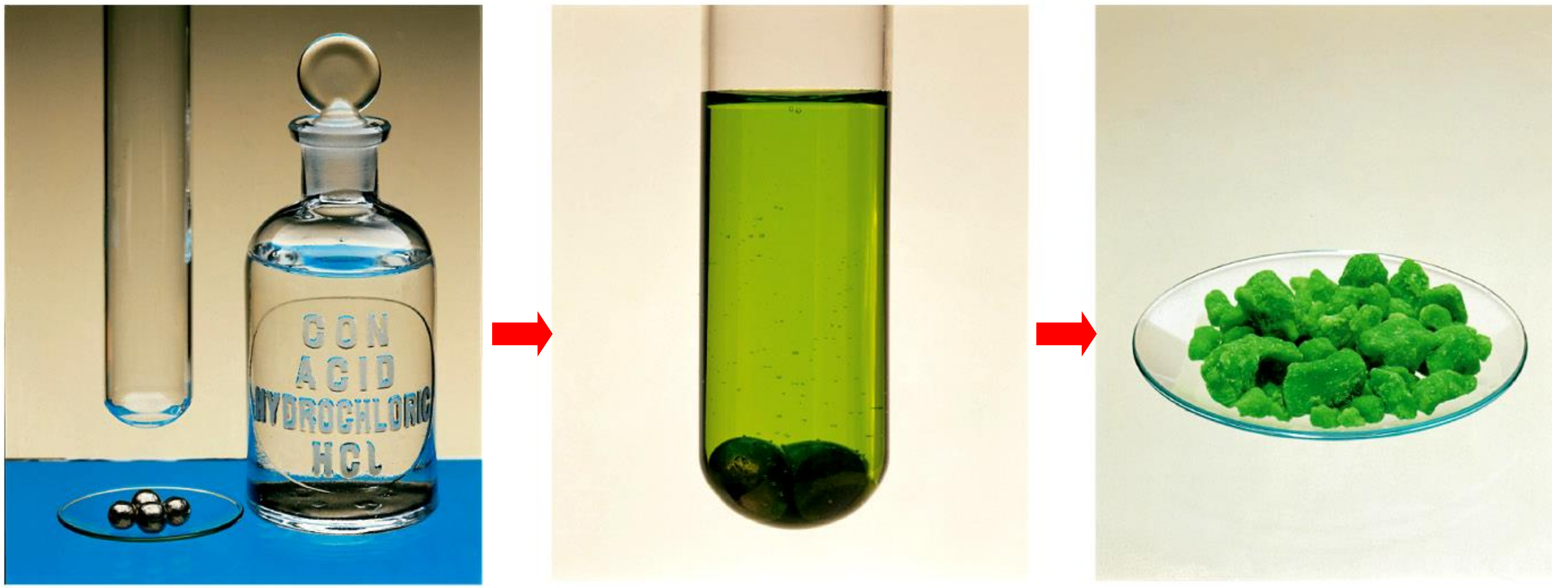


Endothermic solution process

For example, the dissolution of ammonium nitrate (NH_4NO_3) in water, heat is absorbed, not released.

Ammonium nitrate instant ice pack

Beware!



Nickel metal and hydrochloric acid

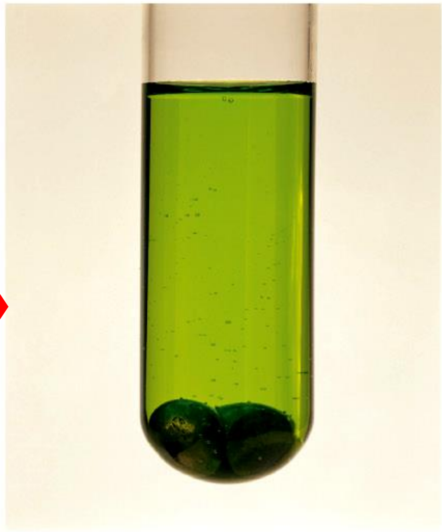
Nickel reacts with hydrochloric acid, forming $NiCl_2(aq)$ and $H_2(g)$. The solution is of $NiCl_2$, not Ni metal

$NiCl_2 \cdot 6H_2O(s)$ remains when solvent evaporated

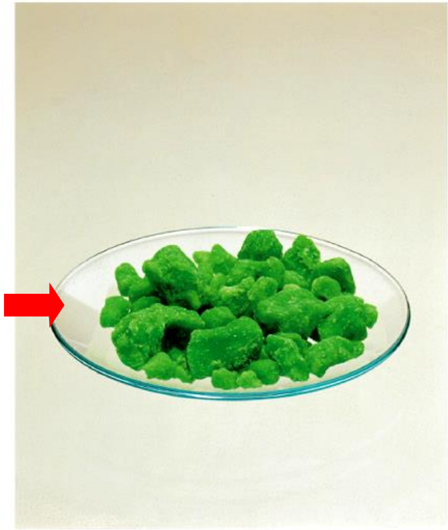
Just because a substance disappears when it comes in contact with a solvent, it doesn't mean the substance dissolved. It may have reacted (Chemical reaction).

Beware!

Nickel metal and hydrochloric acid



Nickel reacts with hydrochloric acid, forming $NiCl_2(aq)$ and $H_2(g)$. The solution is of $NiCl_2$, not Ni metal



$NiCl_2{\cdot}6H_2O(s)$ remains when solvent evaporated

- Dissolution is a physical change—you can get back the original solute by evaporating the solvent.
- If you can't get it back, the substance didn't dissolve, it reacted.

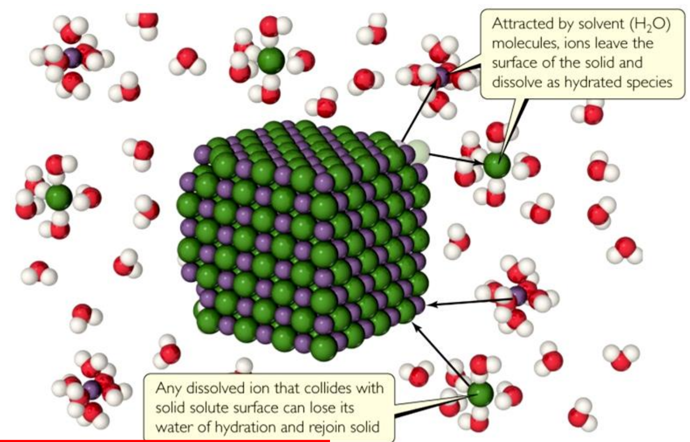
$CuSO_4$ in H_2O?

Types of Solutions

In the process of dissolution, solid solute dissolved in a solvent → concentration of solute particles in solution increase → Increasing the chances that solute particles collide with solid surface and reattach,

i.e. opposite of dissolution process, called crystallization

Dissolved solute is in dynamic equilibrium with solid solute particles



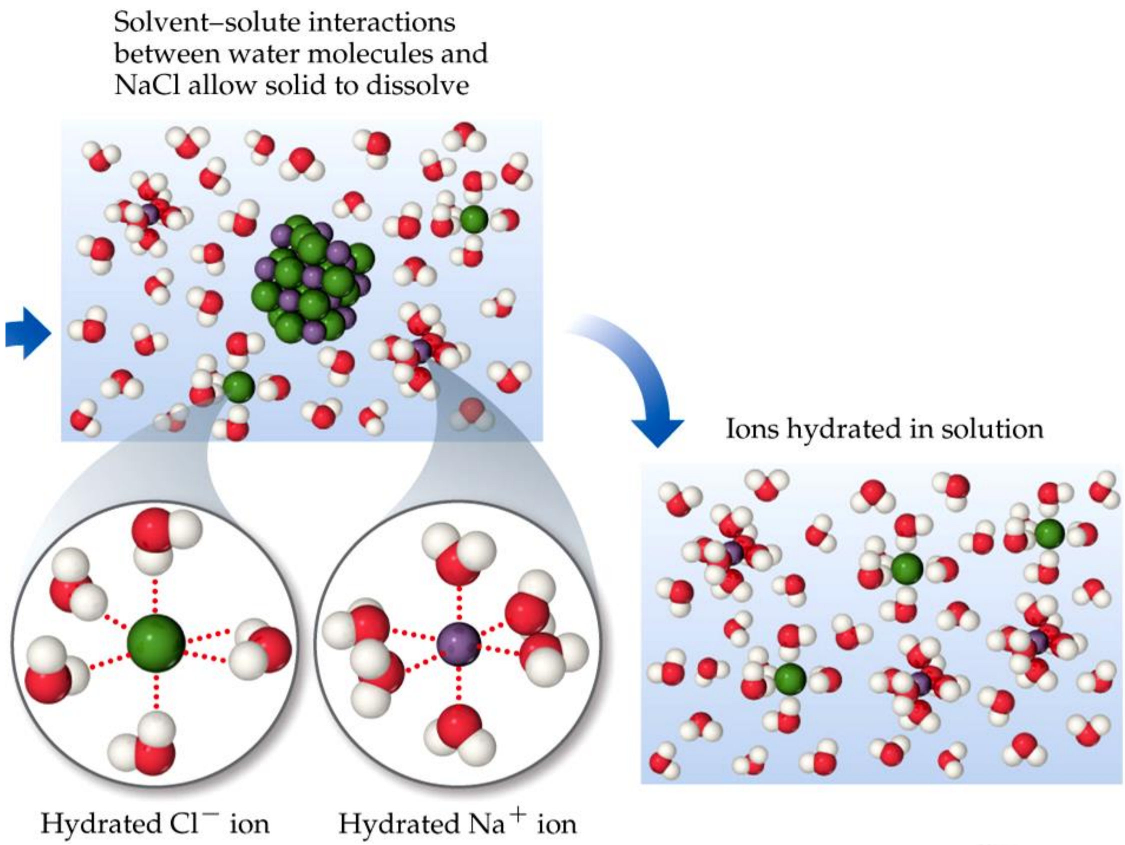
$$\boxed{\text{Solute} + \text{solvent} \underset{\text{crystallize}}{\overset{\text{dissolve}}{\rightleftharpoons}} \text{solution}}$$

In a **saturated** (**饱和**) solution, the solvent holds as much solute as is possible at that temperature.

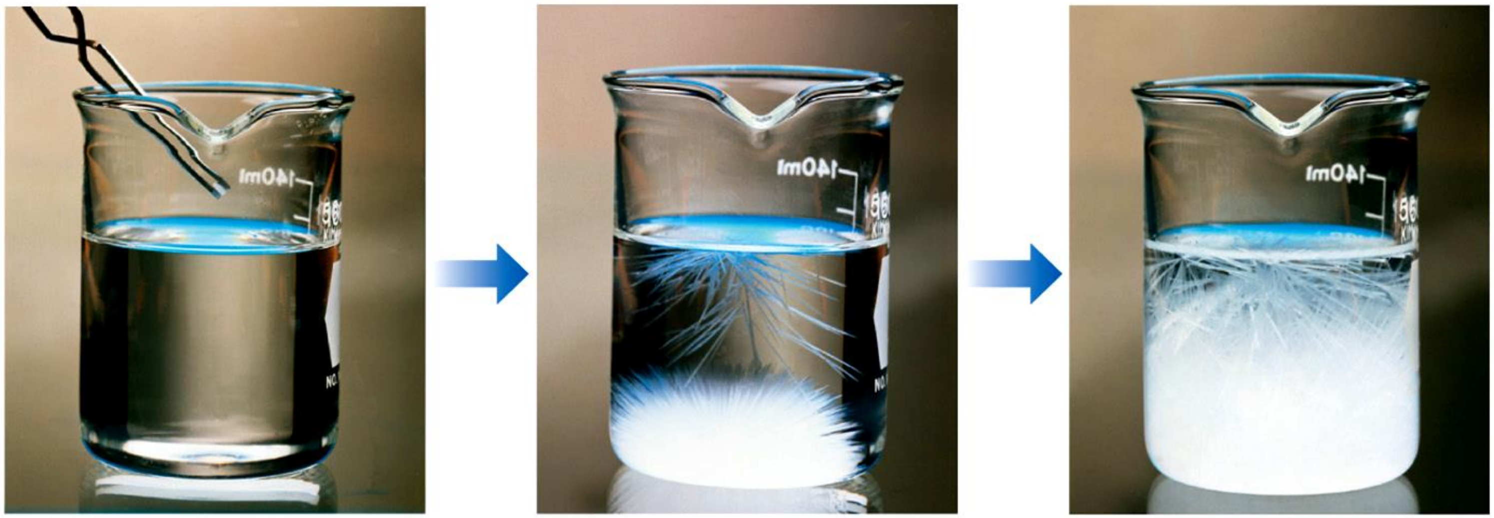
Solubility g/(100 mL) g/(100 g)

Types of Solutions

If we dissolve **less solute than the amount needed** to form a saturated solution, the solution is **unsaturated**.



Types of Solutions



Seed crystal of sodium acetate added to supersaturated solution

Excess sodium acetate crystallizes from solution

Solution arrives at saturation

- In **supersaturated** solutions, the solvent holds more solute than is normally possible at that temperature.
- These solutions are unstable; crystallization can usually be stimulated by adding a "seed crystal" or scratching the side of the flask.

Factors Affecting Solubility

Chemists use the concept "like dissolves like."

– Polar substances tend to dissolve in polar solvents.

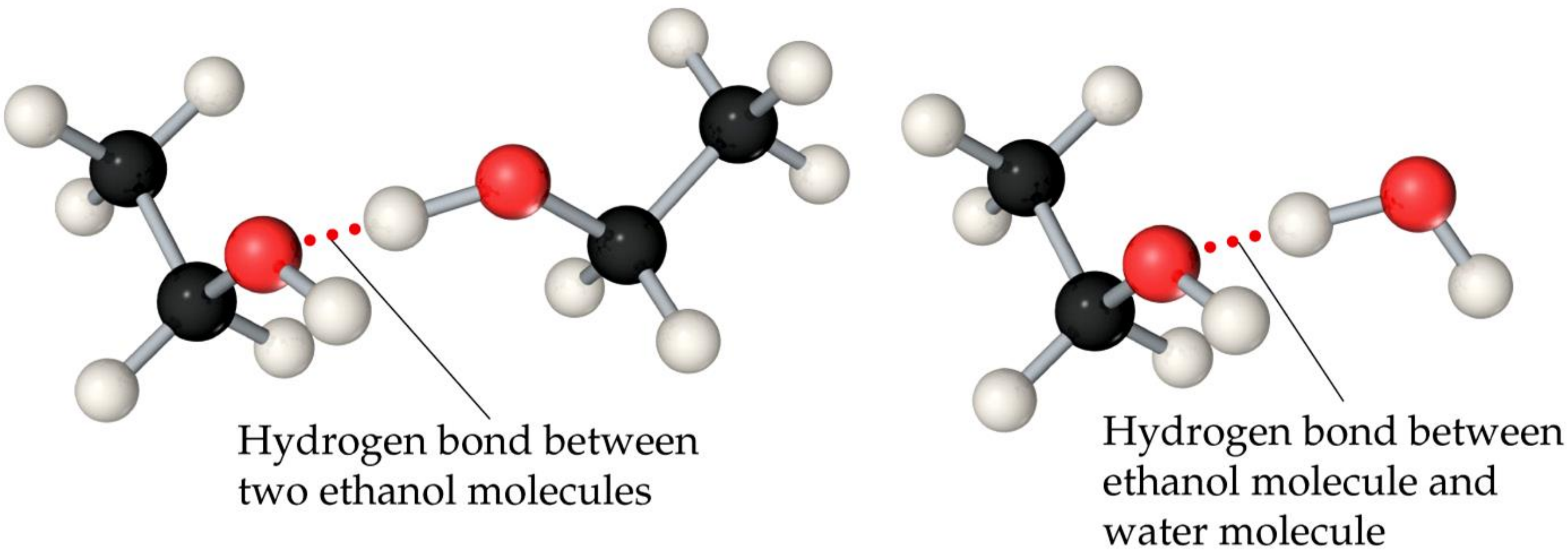
– Nonpolar substances tend to dissolve in nonpolar solvents.

TABLE 13.2 • Solubilities of Some Alcohols in Water and in Hexane*

Alcohol	Solubility in H_2O	Solubility in C_6H_{14}
CH_3OH (methanol)	∞	0.12
CH_3CH_2OH (ethanol)	∞	∞
$CH_3CH_2CH_2OH$ (propanol)	∞	∞
$CH_3CH_2CH_2CH_2OH$ (butanol)	0.11	∞
$CH_3CH_2CH_2CH_2CH_2OH$ (pentanol)	0.030	∞
$CH_3CH_2CH_2CH_2CH_2CH_2OH$ (hexanol)	0.0058	∞

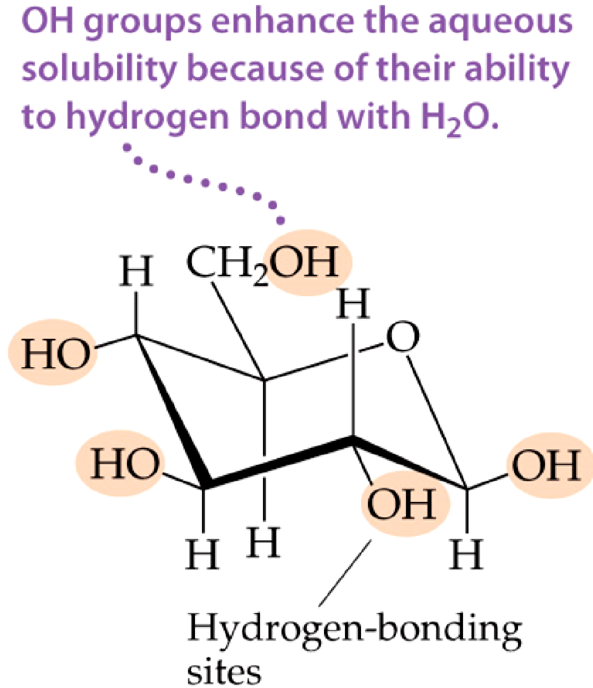
*Expressed in mol alcohol/100 g solvent at 20 °C. The infinity symbol (∞) indicates that the alcohol is completely miscible with the solvent.

Factors Affecting Solubility

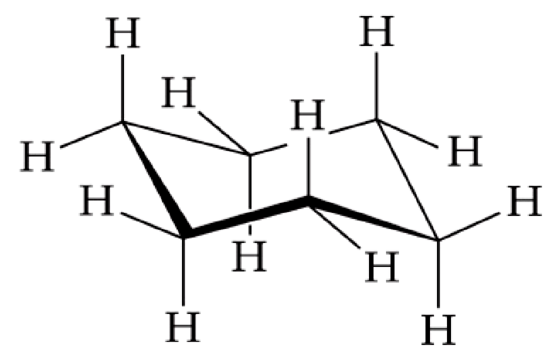


The more similar the intermolecular attractions, the more likely one substance is to be soluble in another.

Factors Affecting Solubility



Glucose, $C_6H_{12}O_6$, has five OH groups and is highly soluble in water

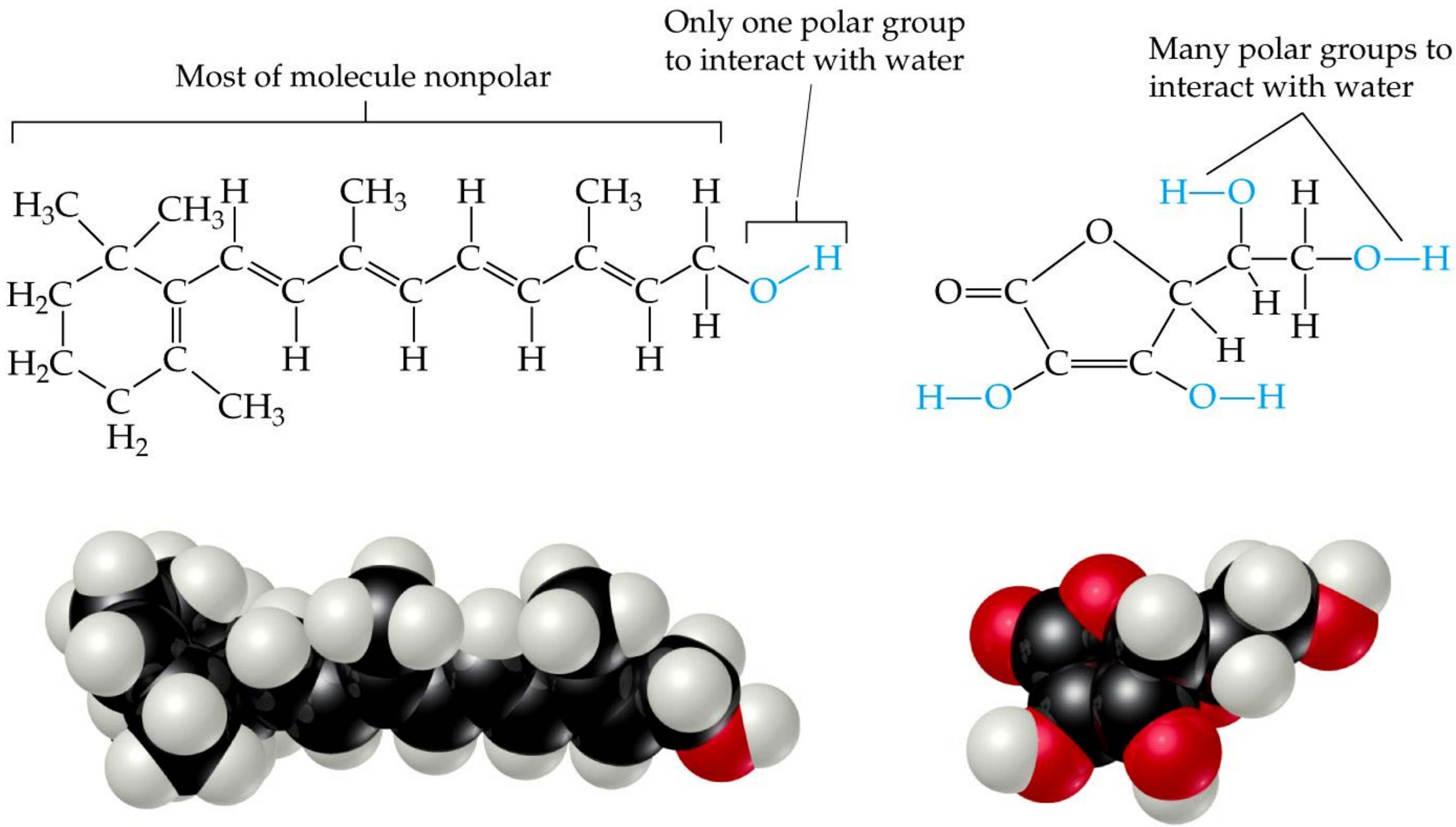


Cyclohexane, C_6H_{12}, which has no polar OH groups, is essentially insoluble in water

葡萄糖

Glucose (which has hydrogen bonding) is very soluble in water, while cyclohexane (which only has dispersion forces) is not soluble in water.

Factors Affecting Solubility



Vitamin A

Vitamin C

- Vitamin A is soluble in nonpolar solvents (like fats).
- Vitamin C is soluble in polar solvents, e.g. water.

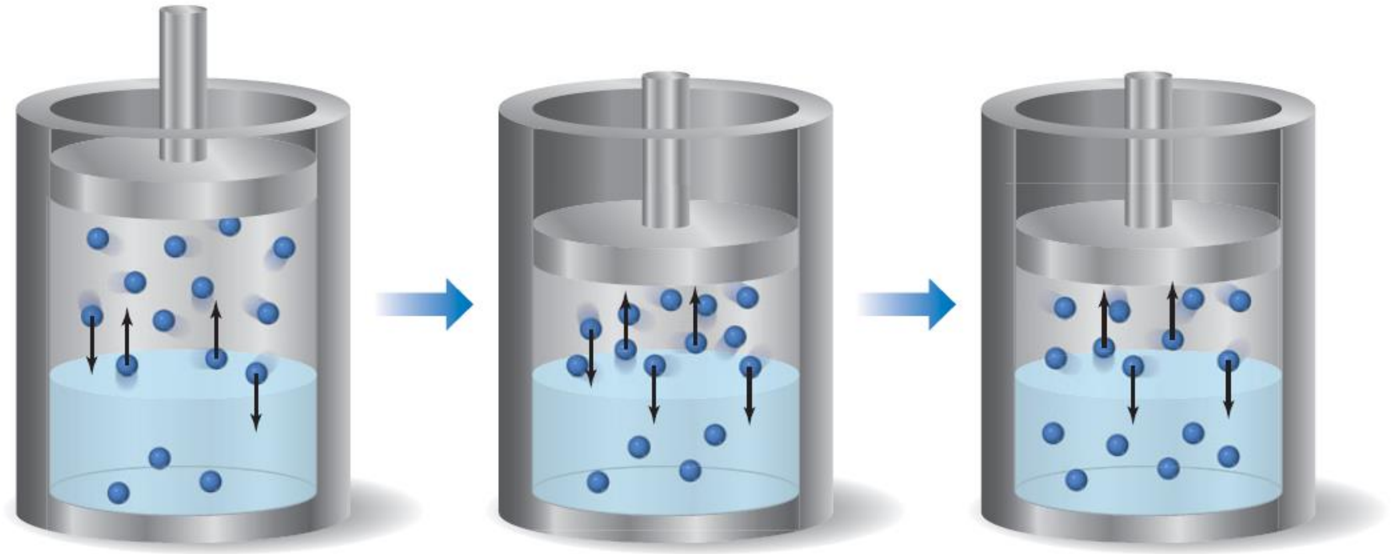
SAMPLE EXERCISE 13.1 Predicting Solubility Patterns

Predict whether each of the following substances is more likely to dissolve in the nonpolar solvent carbon tetrachloride (CCl_4) or in water: C_7H_{16}, Na_2SO_4, HCl, and I_2.

Solubility-Pressure effect

GO FIGURE

If the partial pressure of a gas over a solution is doubled, how has the concentration of gas in the solution changed after equilibrium is restored?



▲ FIGURE 13.14 **Effect of pressure on gas solubility.**

Factors Affecting Solubility-Pressure effect

Gases in Solution

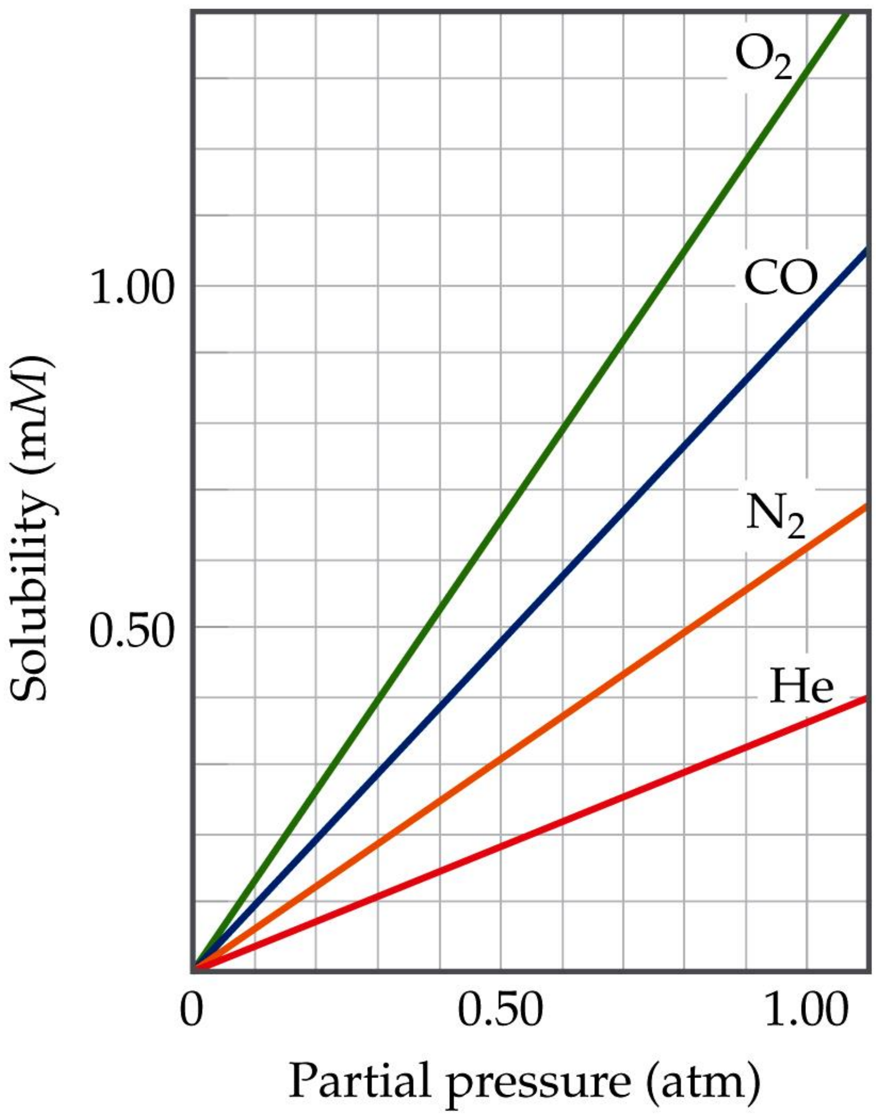
- In general, the solubility of gases in water increases with increasing mass.
- Larger molecules have stronger dispersion forces (higher polarizability).

TABLE 13.1 • Solubilities of Gases in Water at 20 °C, with 1 atm Gas Pressure

Gas	Solubility (M)
N_2	0.69×10^{-3}
CO	1.04×10^{-3}
O_2	1.38×10^{-3}
Ar	1.50×10^{-3}
Kr	2.79×10^{-3}

Increasing mass

Gases in Solution



- The solubility of liquids and solids does not change appreciably with pressure.
- But the solubility of a gas in a liquid is directly proportional to **<u>its pressure</u>**.

Solubility $\propto$ *Partial pressure*

Henry’s Law

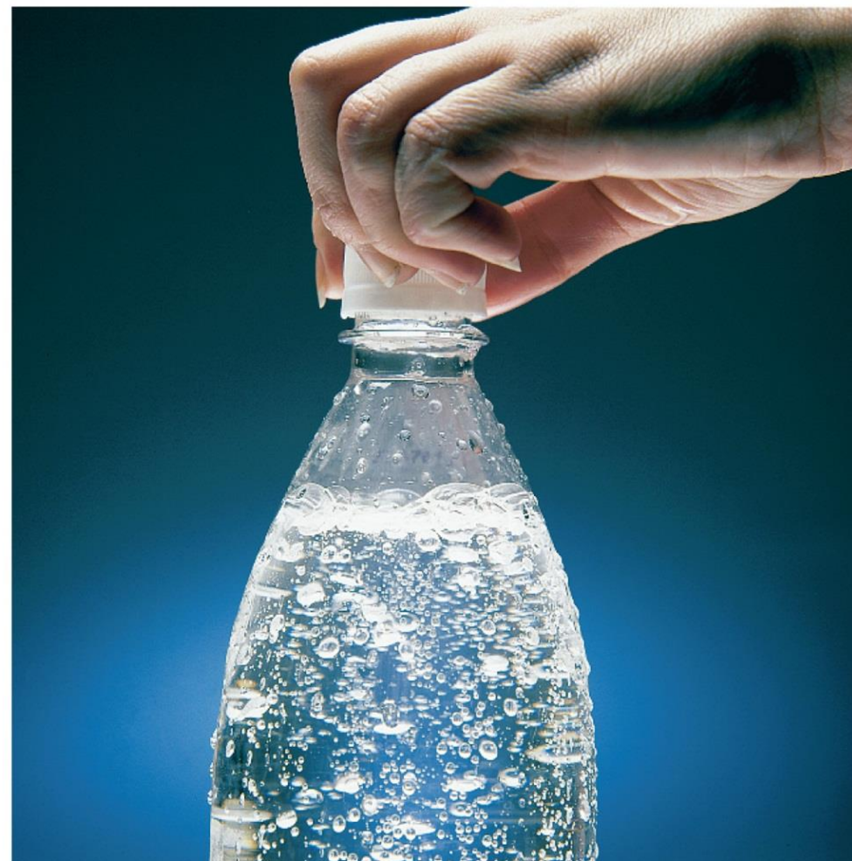
the solubility of a gas in a liquid solvent increases in direct proportion to the partial pressure of the gas above the solution

亨利定律(Henry's law)：“在等温等压下，某种气体在溶液中的溶解度与液面上该气体的平衡压力成正比。”这一定律对于稀溶液中挥发性溶质也同样有用。

$$\boxed{S_g = kP_g}$$

where

- S_g is the solubility of the gas (unit: molarity),
- k is the Henry’s Law constant for that gas in that solvent, and
- P_g is the partial pressure of the gas above the liquid.



Exercise A Henry's Law Calculation

Calculate the concentration of CO_2 in a soft drink that is bottled with a partial pressure of CO_2 of 4.0 atm over the liquid at 25 °C. The Henry's law constant for CO_2 in water at this temperature is 3.4×10^{-2} mol/L-atm.

Solution

Plan: With the information given, we can use Henry's law to calculate the solubility, S_{CO_2}.

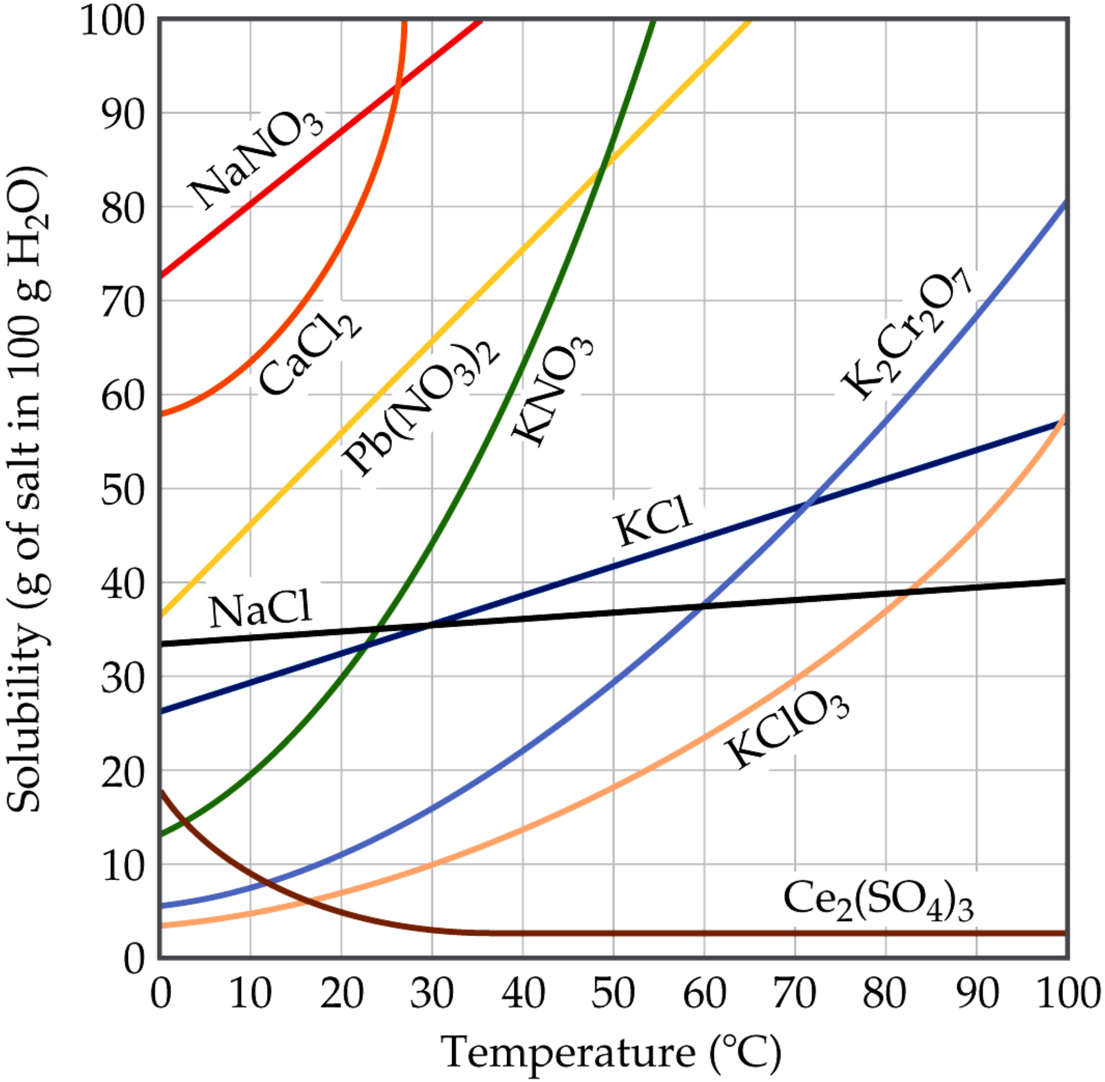
$$S_g = kP_g$$

Solve

$$S_{CO_2} = kP_{CO_2} = (3.4 \times 10^{-2}\ \text{mol/L-atm})(4.0\ \text{atm}) = 0.14\ \text{mol/L} = 0.14\ M$$

Check The units are correct for solubility, and the answer has two significant figures consistent with both the partial pressure of CO_2 and the value of Henry's constant.

Temperature

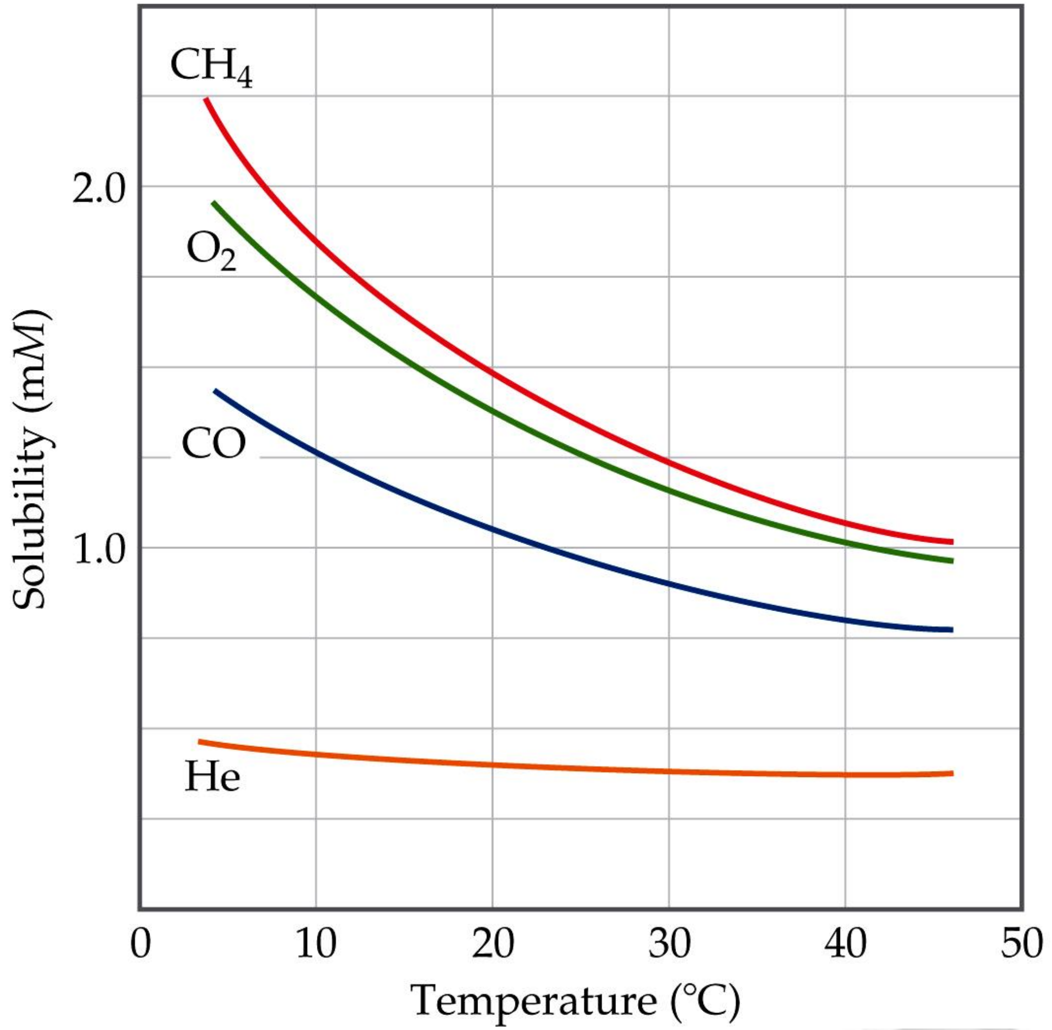


Generally, the solubility of solid solutes in liquid solvents increases with increasing temperature.

Temperature

The opposite is true of gases.

- Solubility decreases with increasing temp.
- Carbonated soft drinks are more “bubbly” if stored in refrigerator.
- Warm lakes have less O_2 dissolved in them than cool lakes.



Ways of Expressing Concentrations of Solutions

Mass Percentage

$$\text{Mass \% of } A = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 100$$

Parts per Million and Parts per Billion

Parts per million (百万分之一) (ppm)

$$\text{ppm} = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 10^6$$

Parts per billion (十亿分之一) (ppb)

$$\text{ppb} = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 10^9$$

Mole Fraction (X)
(摩尔分数)

$$X_A = \frac{\text{moles of } A}{\text{total moles of all components}}$$

In some applications, one needs the mole fraction of *solvent*, not solute—make sure you find the quantity you need!

Molarity (*M*)
(摩尔浓度)

英 [məʊ'lærɪtɪ]

$$M = \frac{\text{moles of solute}}{\text{Volume (liters) of solution}}$$

You will recall this concentration measure from Chapter 4.

Since volume is temperature-dependent, molarity can change with temperature.

Molality (m)
(重量摩尔浓度)

英 [məʊ'lælɪtɪ]

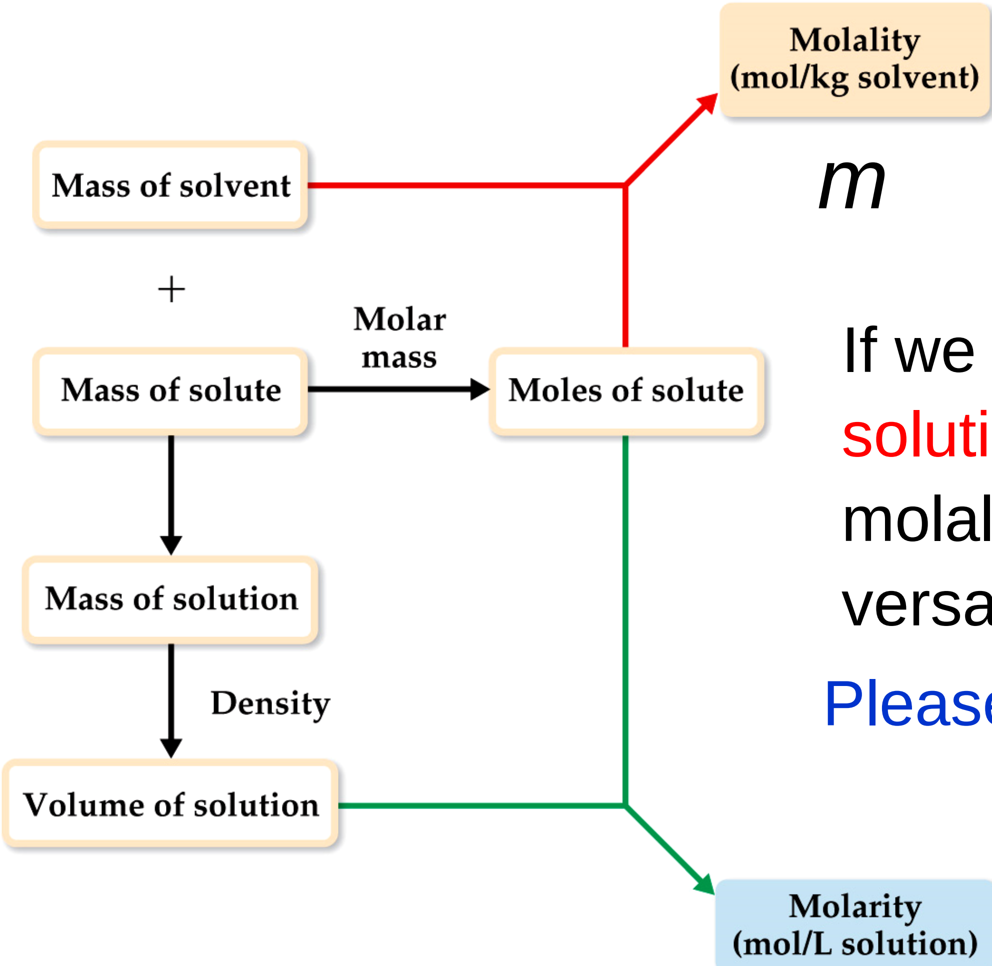
$$m = \frac{\text{moles of solute}}{\text{Mass (kilograms) of solvent}}$$

Since both moles and mass do not change with temperature, molality (unlike molarity) is *not* temperature-dependent,

(temperature-independent).

当涉及温度变化时，常用重量摩尔浓度

Changing Molarity to Molality



m = $\dfrac{\text{moles of solute}}{\text{Mass (kilograms) of solvent}}$

If we know the density of the solution, we can calculate the molality from the molarity, and vice versa.

Please see Sample Exercise 13.6

M = $\dfrac{\text{moles of solute}}{\text{Volume (liters) of solution}}$

$$M = \frac{1000\rho \text{ x mass\%}}{M}$$

$M_{con.HCl}$ = 11.6 M

$M_{con.H2SO4}$ = 18.4 M

SAMPLE EXERCISE 13.3 Calculation of Mass-Related Concentrations

(a) A solution is made by dissolving 13.5 g of glucose ($C_6H_{12}O_6$) in 0.100 kg of water. What is the mass percentage of solute in this solution?

(b) A 2.5-g sample of groundwater was found to contain 5.4 μg of Zn^{2+}. What is the concentration of Zn^{2+} in parts per million?

SAMPLE EXERCISE 13.4 Calculation of Molality

A solution is made by dissolving 4.35 g of glucose ($C_6H_{12}O_6$) in 25.0 mL of water at 25 °C. Calculate the molality of glucose in the solution. Water has a density of 1.00 g/mL.

SAMPLE EXERCISE 13.5 Calculation of Mole Fraction and Molality

An aqueous solution of hydrochloric acid contains 36% HCl by mass. (**a**) Calculate the mole fraction of HCl in the solution. (**b**) Calculate the molality of HCl in the solution.

SAMPLE EXERCISE 13.6 Calculation of Molarity Using the Density of the Solution

A solution with a density of 0.876 g/mL contains 5.0 g of toluene (C_7H_8) and 225 g of benzene. Calculate the molarity of the solution.

Colligative Properties (依数性)

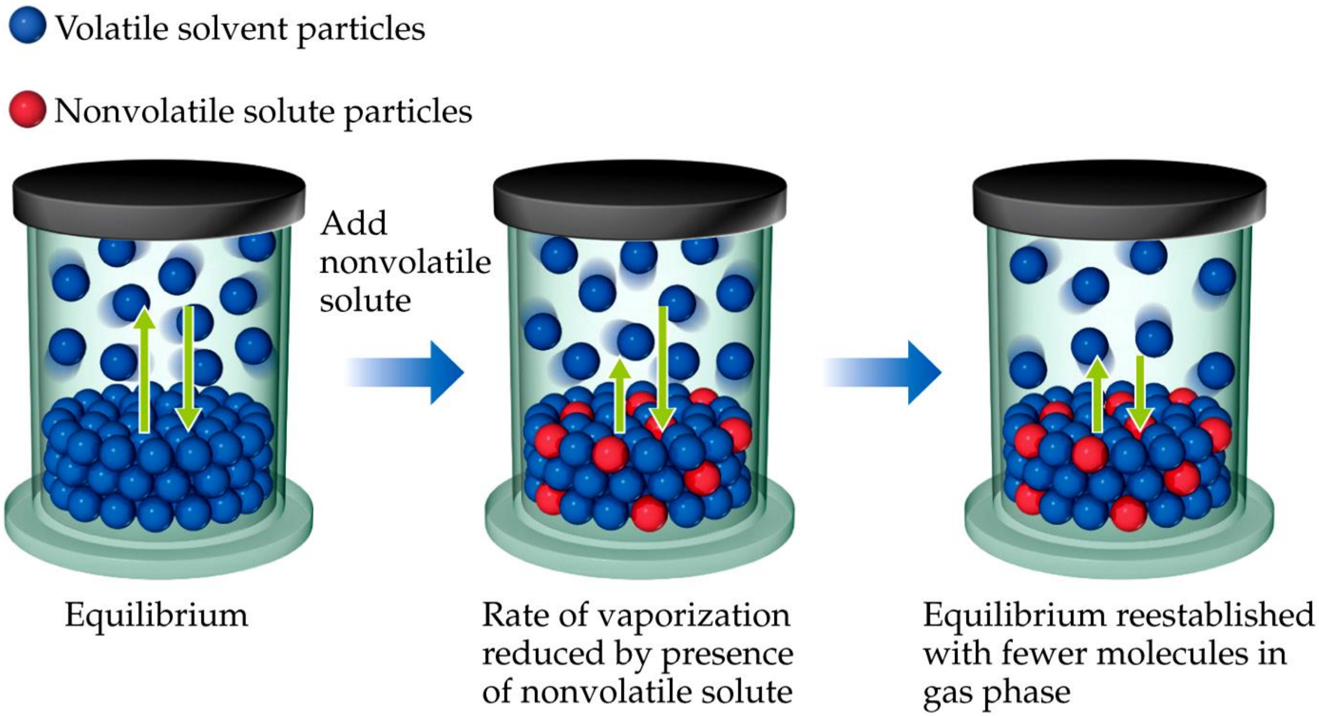
- Addition of solute into a solvent would lower freezing point and raise boiling point
- These properties depends only on the quantity (number) of solute particles present, not on the identity of the solute particles, called **colligative properties**.

稀溶液中溶剂的蒸气压下降、凝固点降低、沸点升高和渗透压的数值，只与溶液中溶质的量有关，与溶质的本性无关，故称这些性质为稀溶液的依数性 (溶液的通性)

- Among colligative properties are
  - Lowering of vapor-pressure
  - Raising of boiling-point
  - Lowering of melting-point
  - Osmotic pressure [ɒz‘mɒtɪk] 渗透压

Solutions

Lowering Vapor Pressure



Because of solute–solvent intermolecular attraction, higher concentrations of nonvolatile solutes make the solvent harder to escape to the vapor phase.

Therefore, the vapor pressure of a solution is lower than that of the pure solvent.

Raoult's Law

$$\boxed{P_{solution} = X_A P^\circ_A + X_B P^\circ_B}$$

拉乌尔定律 (Raoult's Law): 理想溶液在一固定温度下，每一组元的蒸气分压与溶液内各该组元的摩尔分数成正比.

–X is the mole fraction, and

–P° is the normal vapor pressure at that temperature.

If A or B is solid, $P^\circ = 0$

$$\boxed{P_{solution} = X_{solvent} P^\circ_{solvent}}$$

(含有非挥发性溶质的稀溶液) 拉乌尔定律: "在某一温度下，稀溶液的蒸气压等于纯溶剂的蒸气压乘以溶剂的摩尔分数"。

$$\Delta P_{solution} = X_{solute} P^\circ_{solvent}$$

Solutions

(The vapor-pressure lowering is directly proportional to the mole fraction of the solute)

Ideal solution

is defined as one that obeys Raoult's law

The molecules in an ideal solution all influence one another in the same way

- **The solute concentration is <u>low and dilute</u>**
- **Solvent and Solute have similar molecular sizes**
- **Take part in similar types of intermolecular attractions.**

$$P_{solution} = X_{solvent}P^{\circ}_{solvent} \quad [13.10]$$

For example, the vapor pressure of pure water at 20 °C is $P^{\circ}_{H_2O} = 17.5$ torr. Imagine holding the temperature constant while adding glucose ($C_6H_{12}O_6$) to the water so that the mole fractions in the resulting solution are $X_{H_2O} = 0.800$ and $X_{C_6H_{12}O_6} = 0.200$. According to Equation 13.10, the vapor pressure of the water above this solution is 80.0% of that of pure water:

$$P_{solution} = (0.800)(17.5 \text{ torr}) = 14.0 \text{ torr}$$

The presence of the *nonvolatile* solute lowers the vapor pressure of the *volatile* solvent by 17.5 torr − 14.0 torr = 3.5 torr.

The vapor-pressure lowering, ΔP, is directly proportional to the mole fraction of the solute, X_{solute}:

$$\Delta P = X_{solute}\, P^{\circ}_{solvent} \quad [13.11]$$

Thus, for the example of the solution of glucose in water, we have

$$\Delta P = X_{C_6H_{12}O_6}\, P^{\circ}_{H_2O} = (0.200)(17.5 \text{ torr}) = 3.50 \text{ torr}$$

Solutions sometimes have two or more volatile components. Gasoline, for example, is a solution of several volatile liquids. To gain some understanding of such mixtures, consider an ideal solution of two volatile liquids, A and B. (For our purposes here, it does not matter which we call the solute and which the solvent.) The partial pressures above the solution are given by Raoult's law:

$$P_A = X_A P_A^\circ \text{ and } P_B = X_B P_B^\circ$$

and the total vapor pressure above the solution is

$$P_{total} = P_A + P_B = X_A P_A^\circ + X_B P_B^\circ$$

Consider a mixture of 1.0 mol of benzene (C_6H_6) and 2.0 mol of toluene (C_7H_8) ($X_{ben} = 0.33$, $X_{tol} = 0.67$). At 20 °C, the vapor pressures of the pure substances are $P_{ben}^\circ = 75$ torr and $P_{tol}^\circ = 22$ torr. Thus, the partial pressures above the solution are

$$P_{ben} = (0.33)(75 \text{ torr}) = 25 \text{ torr}$$
$$P_{tol} = (0.67)(22 \text{ torr}) = 15 \text{ torr}$$

and the total vapor pressure above the liquid is

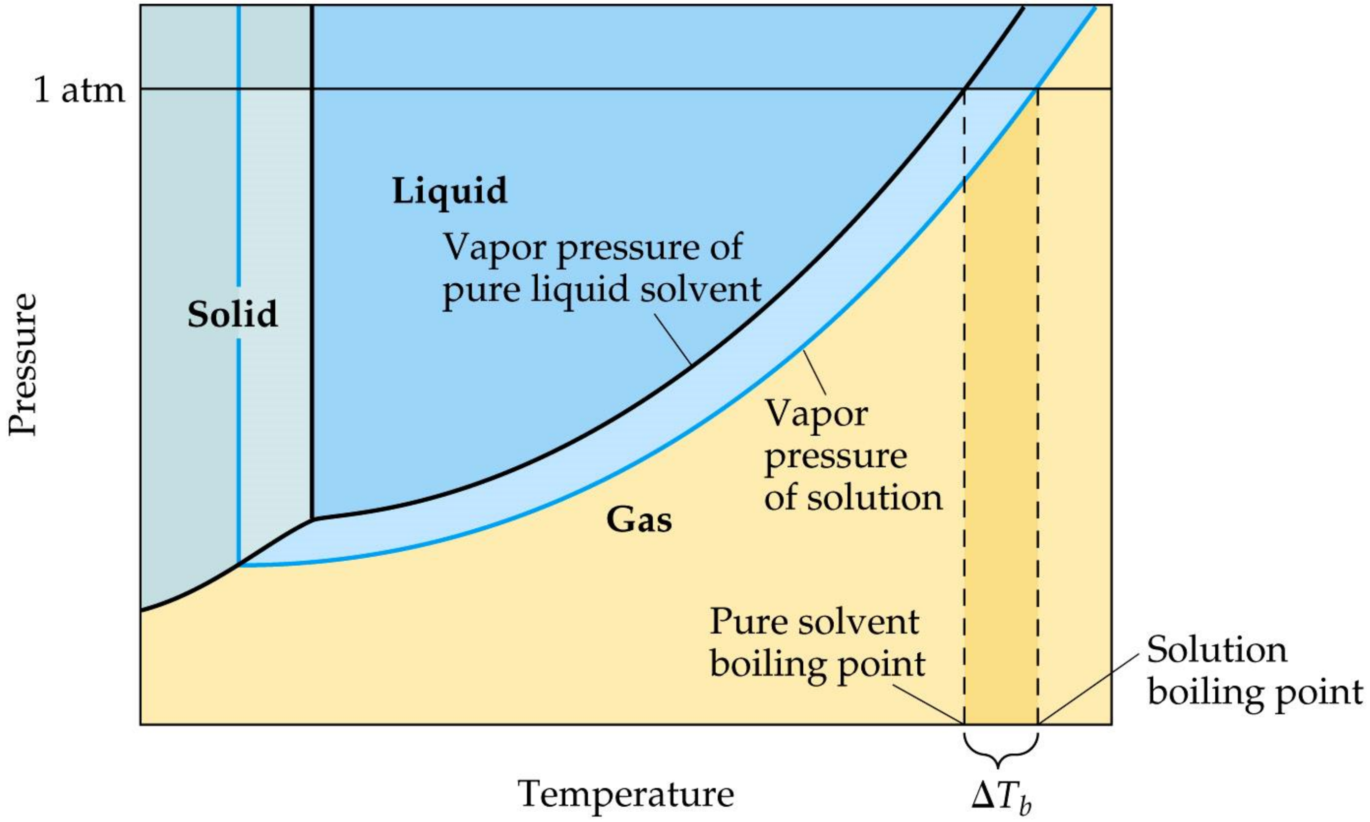
$$P_{total} = P_{ben} + P_{tol} = 25 \text{ torr} + 15 \text{ torr} = 40 \text{ torr}$$

Note that the vapor is richer in benzene, the more volatile component.

SAMPLE EXERCISE 13.7 Calculation of Vapor Pressure of a Solution

Glycerin ($C_3H_8O_3$) is a nonvolatile nonelectrolyte with a density of 1.26 g/mL at 25 °C. Calculate the vapor pressure at 25 °C of a solution made by adding 50.0 mL of glycerin to 500.0 mL of water. The vapor pressure of pure water at 25 °C is 23.8 torr (Appendix B), and its density is 1.00 g/mL.

Boiling-Point Elevation and Freezing-Point Depression



Nonvolatile solute–solvent interactions also cause solutions to have higher boiling points and lower freezing points than the pure solvent.

Boiling-Point Elevation

TABLE 13.3 • Molal Boiling-Point-Elevation and Freezing-Point-Depression Constants

Solvent	Normal Boiling Point (°C)	K_b (°C/m)	Normal Freezing Point (°C)	K_f (°C/m)
Water, H_2O	100.0	0.51	0.0	1.86
Benzene, C_6H_6	80.1	2.53	5.5	5.12
Ethanol, C_2H_5OH	78.4	1.22	−114.6	1.99
Carbon tetrachloride, CCl_4	76.8	5.02	−22.3	29.8
Chloroform, $CHCl_3$	61.2	3.63	−63.5	4.68

The change in boiling point is proportional to the molality (m) of the solution:

$$\boxed{\Delta T_b = K_b \cdot m}$$

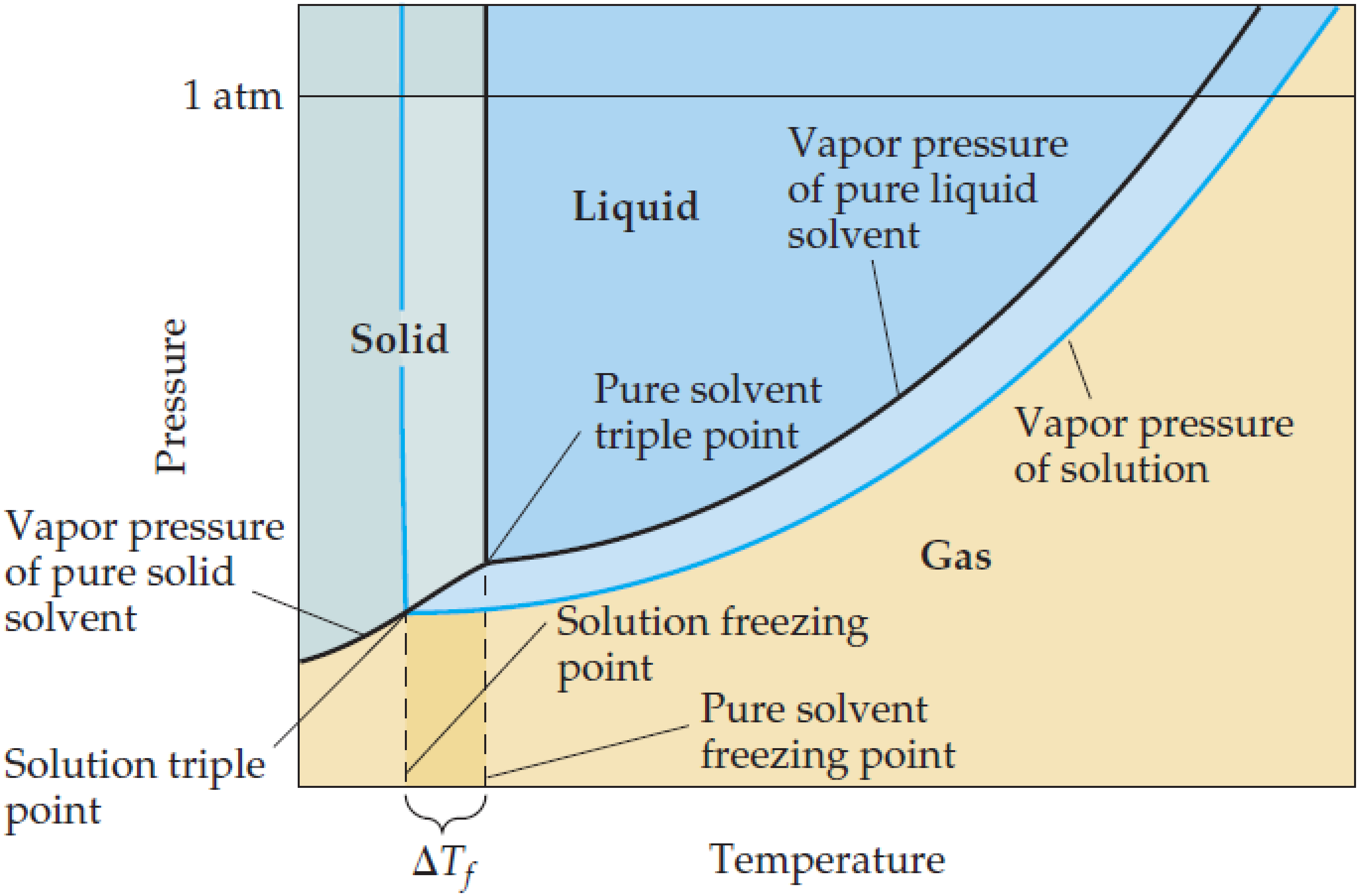
where K_b is the molal boiling-point elevation constant, a property of the solvent.

$$\rightarrow \quad T_b = T^o_b + \Delta T_b$$

Solutions

(T^o_b = normal boiling point)

Freezing-Point Depression



▲ **FIGURE 13.24** **Phase diagram illustrating freezing-point depression.**

Freezing-Point Depression

TABLE 13.3 • Molal Boiling-Point-Elevation and Freezing-Point-Depression Constants

Solvent	Normal Boiling Point (°C)	K_b (°C/m)	Normal Freezing Point (°C)	K_f (°C/m)
Water, H_2O	100.0	0.51	0.0	1.86
Benzene, C_6H_6	80.1	2.53	5.5	5.12
Ethanol, C_2H_5OH	78.4	1.22	−114.6	1.99
Carbon tetrachloride, CCl_4	76.8	5.02	−22.3	29.8
Chloroform, $CHCl_3$	61.2	3.63	−63.5	4.68

The change in freezing point can be found similarly:

$$\boxed{\Delta T_f = K_f \bullet m}$$

Here K_f is the molal freezing-point depression constant of the solvent.

$$T_f = T^o_f - \Delta T_f$$

(T^o_f = normal freezing point)

For water, K_f is 1.86 °C/m. Therefore, any aqueous solution that is 1 m in nonvolatile solute particles (such as 1 m $C_6H_{12}O_6$ or 0.5 m NaCl) freezes at the temperature that is 1.86 °C lower than the freezing point of pure water.

For water, K_f is 1.86 °C/*m*. Therefore, any aqueous solution that is 1 *m* in nonvolatile solute particles (such as 1 *m* $C_6H_{12}O_6$ or 0.5 *m* NaCl) freezes at the temperature that is 1.86 °C lower than the freezing point of pure water.

The freezing-point depression caused by solutes has useful applications: it is why antifreeze works in car cooling systems, and why calcium chloride ($CaCl_2$) promotes the melting of ice on roads during winter.

Antifreeze:

抗冻剂有甲醇、乙醇、乙二醇、水溶性酰胺和氯化钙、盐水等。其中乙二醇的抗冻性能优异，是最主要的抗冻剂。例如：乙二醇含量为40％（质量）的水溶液，冰点为 -24℃；而含乙二醇58％（质量）的水溶液，其冰点为-48℃。世界上需用抗冻剂的系统中，约90％采用乙二醇及其衍生物作抗冻剂。

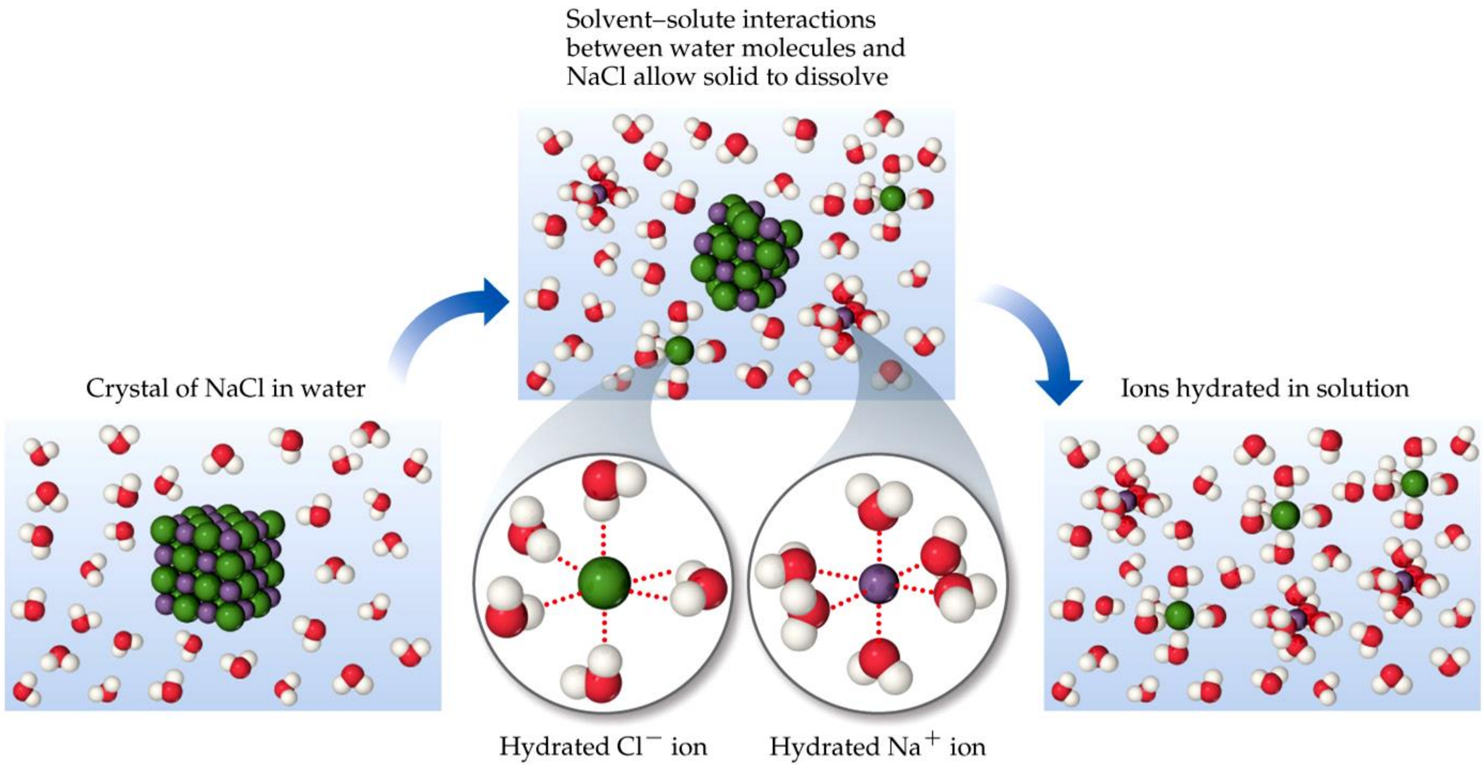
乙二醇 编辑

乙二醇（ethylene glycol）又名"甘醇"、"1,2-亚乙基二醇"，简称EG。化学式为(CH2OH)2，是最简单的二元醇。乙二醇是无色无臭、有甜味液体，对动物有毒性，人类致死剂量约为1.6 g/kg。乙二醇能与水、丙酮互溶，但在醚类中溶解度较小。用作溶剂、防冻剂以及合成涤纶的原料。乙二醇的高聚物聚乙二醇（PEG）是一种相转移催化剂，也用于细胞融合；其硝酸酯是一种炸药。

中文名	乙二醇	水溶性	与水互溶
英文名	Ethylene glycol	密　度	1.1155(20℃)
别　称	甘醇，EG，甘醇型防冻液，MEG	外　观	无色
化学式	(CH2OH)2	闪　点	111.1℃
分子量	62.068	危险性描述	吞食有害
CAS登录号	107-21-1	临界压力	7699KPa
EINECS登录号	203-473-3	临界温度	372℃
熔　点	-12.9℃	偏心因子	0.27
沸　点	197.3℃	临界摩尔体积	186C3/mol [1]

Colligative Properties of Electrolytes (电解质)

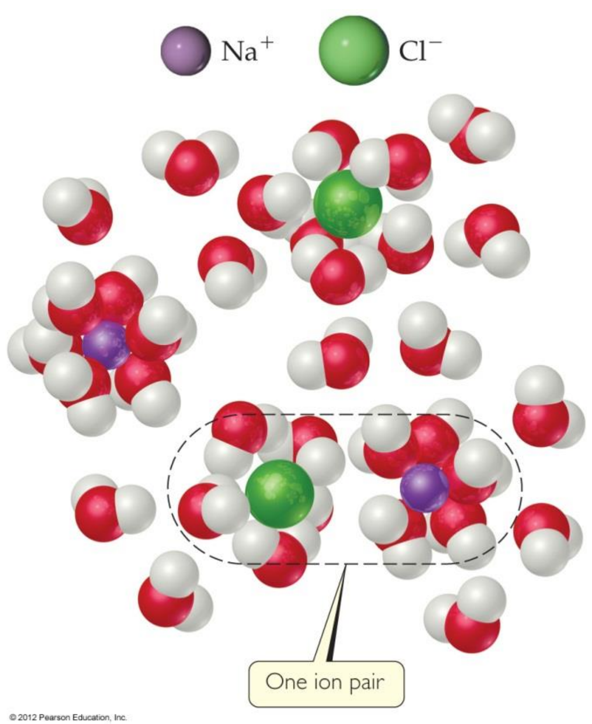
Since the colligative properties of electrolytes depend on the number of particles dissolved, solutions of electrolytes (which dissociate in solution) should show greater changes than those of nonelectrolytes.



Solutions

Colligative Properties of Electrolytes

However, a 1*M* solution of NaCl does not show twice the change in freezing point that a 1*M* solution of methanol does.



One mole of NaCl in water does not really give rise to two moles of ions.

Some Na^+ and Cl^- reassociate for a short time, so the true concentration of particles is somewhat less than two times the concentration of NaCl.

van’t Hoff Factor
范特霍夫

- Reassociation is more likely at higher concentration.
- Therefore, the number of particles present is concentration-dependent.

TABLE 13.4 • van’t Hoff Factors for Several Substances at 25 °C

Compound	Concentration 0.100 *m*	0.0100 *m*	0.00100 *m*	Limiting Value
Sucrose	1.00	1.00	1.00	1.00
NaCl	1.87	1.94	1.97	2.00
K_2SO_4	2.32	2.70	2.84	3.00
$MgSO_4$	1.21	1.53	1.82	2.00

- We modify the previous equations by multiplying by the van’t Hoff factor, *i:*

$$\boxed{\Delta T = K \cdot m \cdot i}$$

Jacobus Henricus van’t Hoff

范特霍夫

Nobel Prize, 1901

$$\boxed{\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent}) = iK_b m} \quad [13.12]$$

In this equation, T_b(solution) is the boiling point of the solution, T_b(solvent) is the boiling point of the pure solvent, m is the molality of the solute, K_b is the **molal boiling-point-elevation constant** for the solvent (which is a proportionality constant that is experimentally determined for each solvent), and i is the van't Hoff factor. For a nonelectrolyte, we can always assume $i = 1$; for an electrolyte, i will depend on how the substance ionizes in that solvent. For instance, $i = 2$ for NaCl in water, assuming complete dissociation of ions. As a result, we expect the boiling-point elevation of a 1 m aqueous solution of NaCl to be twice as large as the boiling-point elevation of a 1 m solution of a nonelectrolyte such as sucrose. Thus, to properly predict the effect of a particular solute on boiling-point elevation (or any other colligative property), it is important to know whether the solute is an electrolyte or a nonelectrolyte. (Sections 4.1 and 4.3)

$$\boxed{\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent}) = -iK_f m} \quad [13.13]$$

The proportionality constant K_f is the **molal freezing-point-depression constant**, analogous to K_b for boiling point elevation. Note that because the solution freezes at a *lower* temperature than does the pure solvent, the value of ΔT_f is *negative.*

Some typical values of K_b and K_f for several common solvents are given in ▲ **Table 13.3**. For water, the table shows $K_b = 0.51\ °C/m$, which means that the boiling point of any aqueous solution that is 1 m in nonvolatile solute particles is 0.51 °C higher than the boiling point of pure water. Because solutions generally do not behave ideally, the constants listed in Table 13.3 serve well only for solutions that are rather dilute.

SAMPLE EXERCISE 13.8 Calculation of Boiling-Point Elevation and Freezing-Point Depression

Automotive antifreeze contains ethylene glycol, $CH_2(OH)CH_2(OH)$, a nonvolatile nonelectrolyte, in water. Calculate the boiling point and freezing point of a 25.0% by mass solution of ethylene glycol in water.

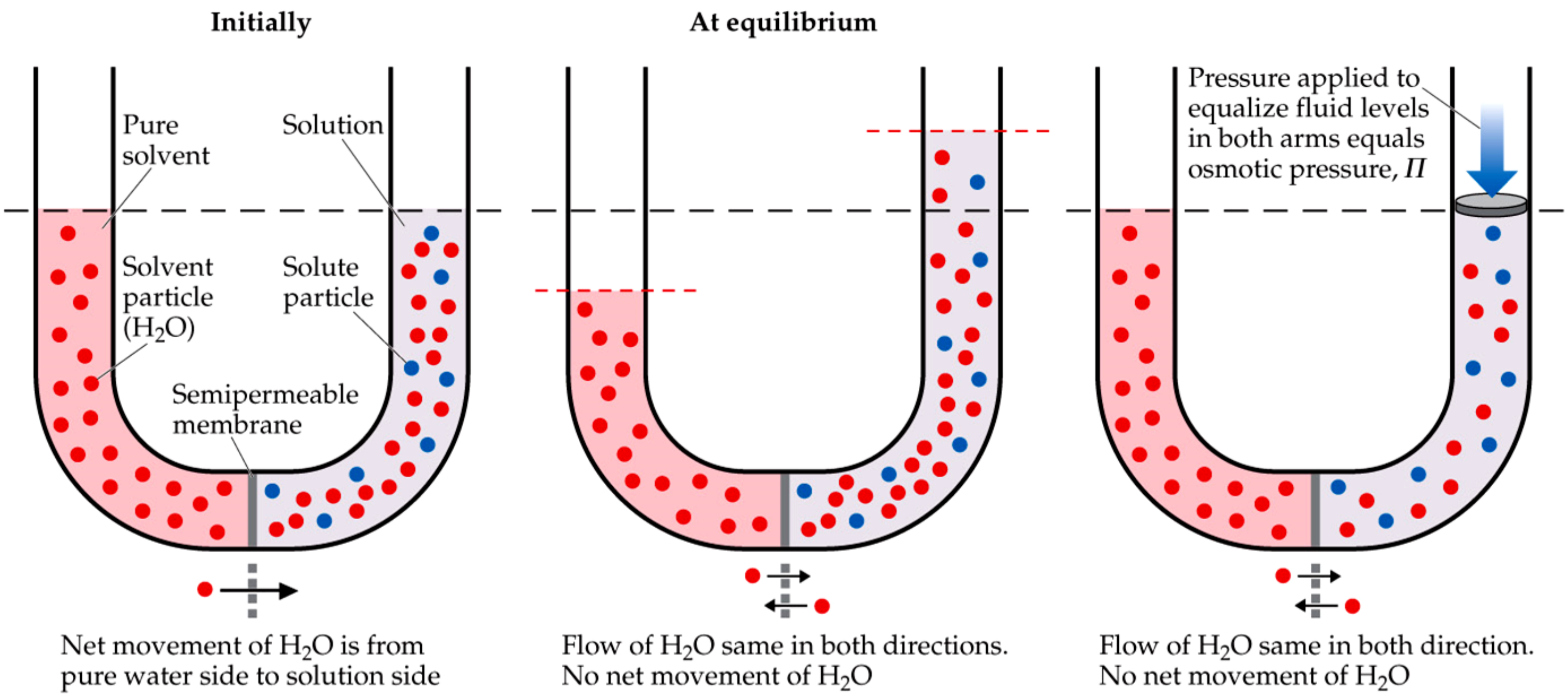
Osmosis (渗透)

[ɒzˈməʊsɪs]

- Some substances form semipermeable (半透性的) membranes（薄膜）, allowing some smaller particles to pass through, but blocking other larger particles.
- In biological systems, most semipermeable membranes allow water to pass through, but not for solutes.

Solutions

Osmosis



In osmosis, there is net movement of solvent from the area of **higher solvent concentration** (*lower solute concentration*) to the area of **lower solvent concentration** (*higher solute concentration*).

Osmotic Pressure

The pressure required to stop osmosis, known as **osmotic pressure**, π, is

$$\boxed{\pi = \left(\frac{n}{V}\right)RT = iMRT}$$

（范特霍夫方程式）

where M is the molarity of the solution，摩尔浓度；

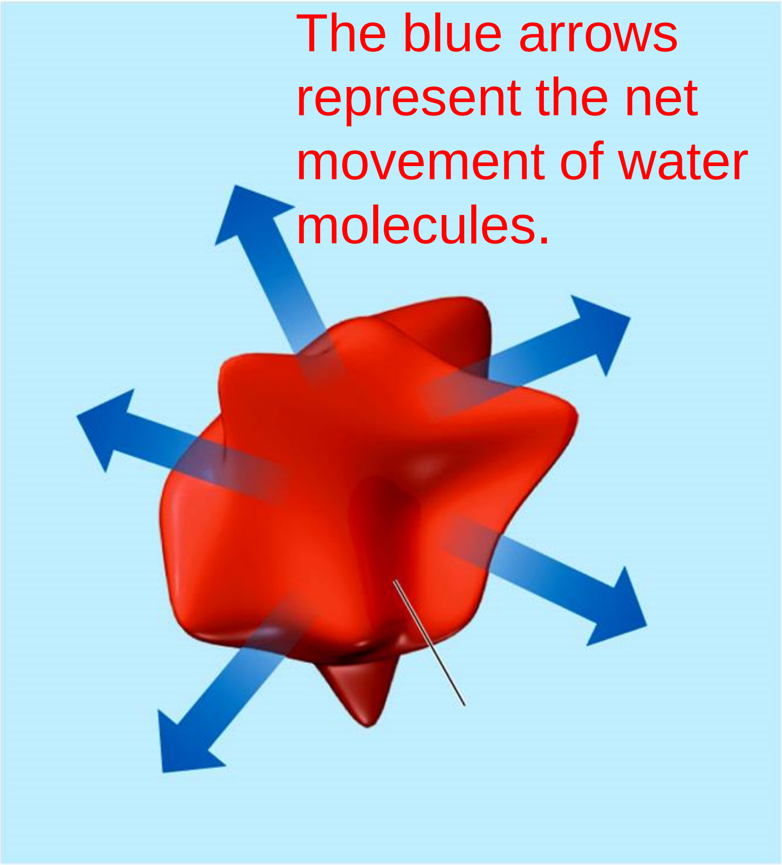
i, the van't Hoff factor .

If the osmotic pressure is the same on both sides of a membrane (i.e., the concentrations are the same), the two solutions are **isotonic (等渗溶液)**.

Osmosis in Blood Cells

- If the solute concentration outside the cell is greater than that inside the cell, the solution is hypertonic (高渗的). [ˌhaɪpə'tɒnɪk]

- Water will flow out of the cell, and crenation (皱缩) results.



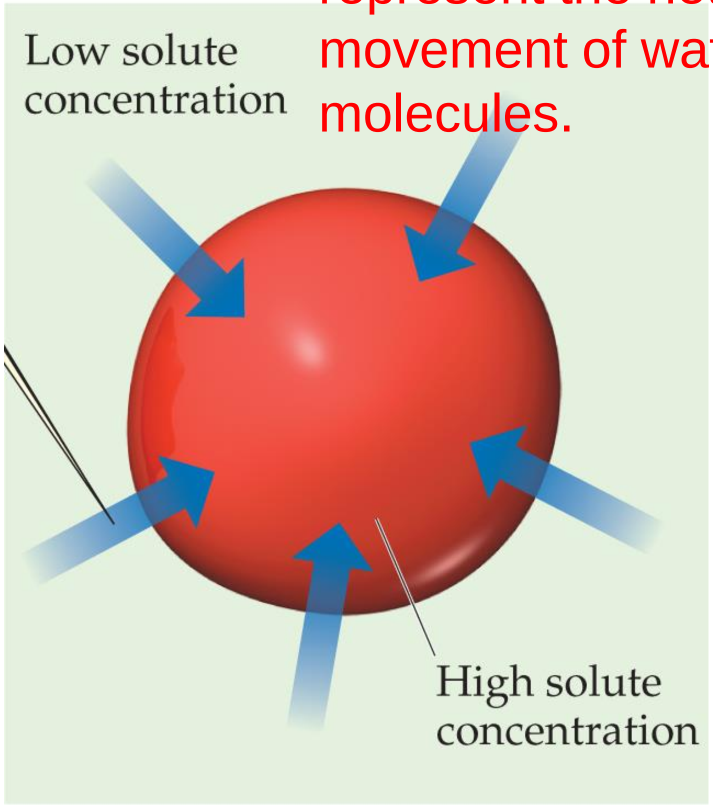
Solutions

Osmosis in Blood Cells

- If the solute concentration outside the cell is less than that inside the cell, the solution is hypotonic (低渗的).   [ˌhaɪpə'tɒnɪk]

- Water will flow into the cell, and hemolysis (溶血) results.

The blue arrows represent the net movement of water molecules.



Hemolysis of red blood cell placed in hypotonic environment

The average osmotic pressure of blood is 7.7 atm at 25 °C.

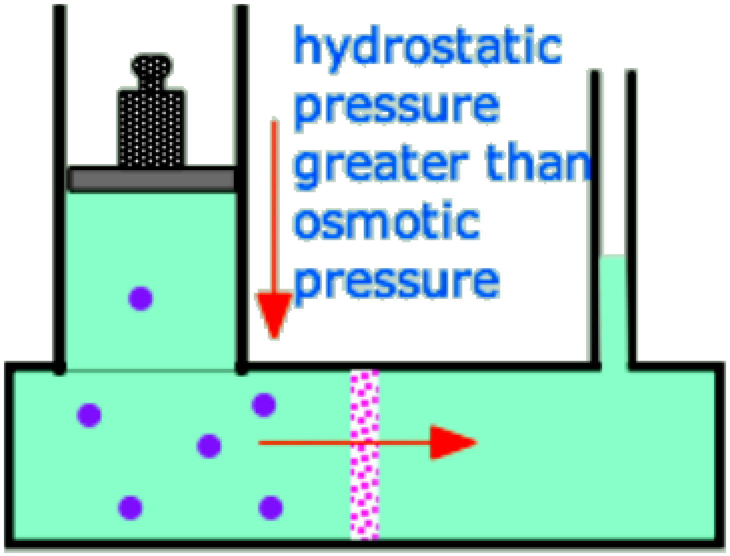
SAMPLE EXERCISE 13.9 Osmotic Pressure Calculations

The average osmotic pressure of blood is 7.7 atm at 25 °C. What molarity of glucose ($C_6H_{12}O_6$) will be isotonic with blood?

SAMPLE EXERCISE 13.10 Molar Mass from Freezing-Point Depression

A solution of an unknown nonvolatile nonelectrolyte was prepared by dissolving 0.250 g of the substance in 40.0 g of CCl_4. The boiling point of the resultant solution was 0.357 °C higher than that of the pure solvent. Calculate the molar mass of the solute.

Reverse Osmosis (反渗)



流体静压

Applying a hydrostatic pressure greater than this to the high-solute side of an osmotic cell will force water to flow back into the fresh-water side. This process is now the major technology employed to desalinate[ˌdi:‘sæləneɪt] ocean water and to reclaim “used” water from power plants, runoff, and even from sewage [ˈsu:ɪdʒ] 污水.

Solutions

Colloids (胶体)

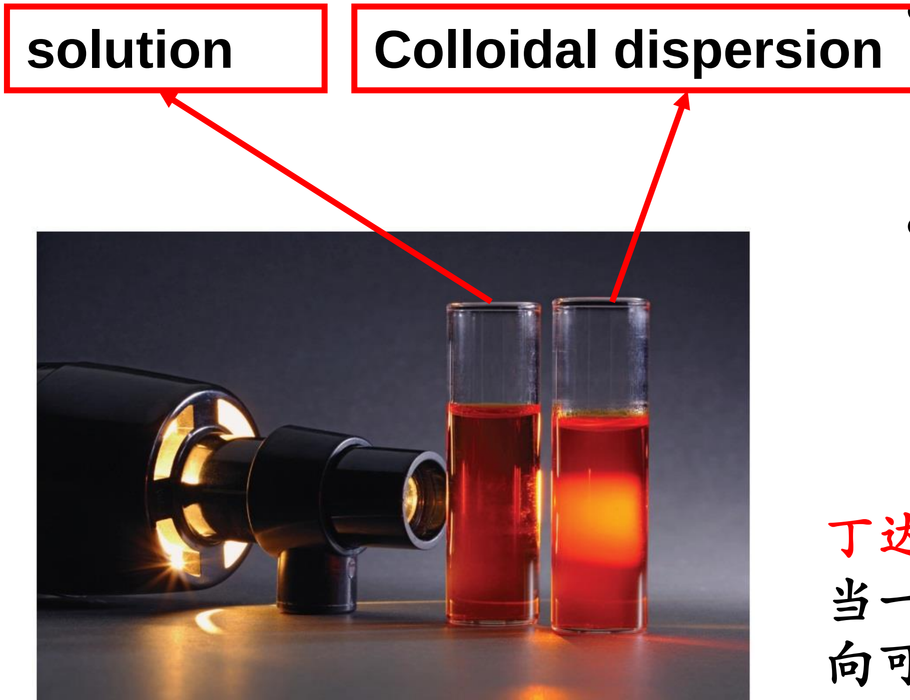
Suspensions (悬浮) of particles larger than individual ions or molecules, but too small to be settled out by gravity, are called **colloids**.

Colloid particles range in diameter from 5 to 1000 nm

TABLE 13.5 • Types of Colloids

Phase of Colloid	Dispersing (solvent-like) Substance	Dispersed (solute-like) Substance	Colloid Type	Example
Gas	Gas	Gas	—	None (all are solutions)
Gas	Gas	Liquid	Aerosol	Fog
Gas	Gas	Solid	Aerosol	Smoke
Liquid	Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Liquid	Emulsion	Milk
Liquid	Liquid	Solid	Sol	Paint
Solid	Solid	Gas	Solid foam	Marshmallow
Solid	Solid	Liquid	Solid emulsion	Butter
Solid	Solid	Solid	Solid sol	Ruby glass

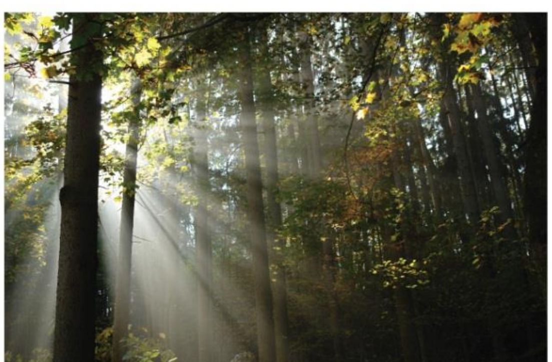
Tyndall Effect (丁达尔效应)



- Colloidal suspensions can scatter rays of light.
- This phenomenon is known as the **Tyndall effect**.

丁达尔效应：

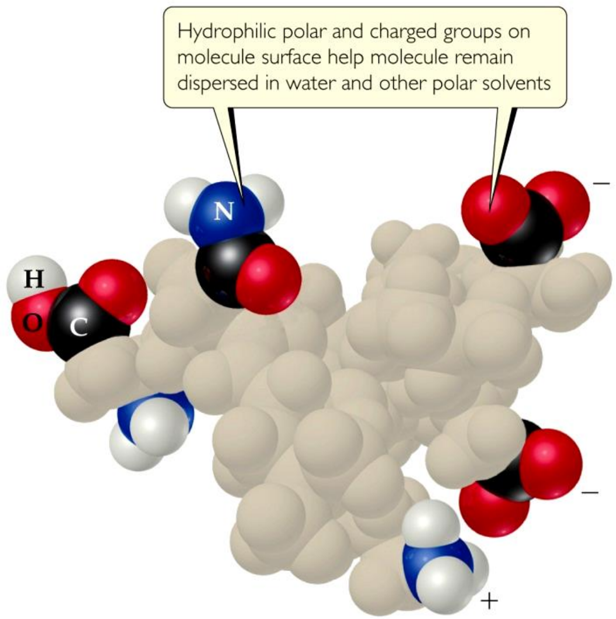
当一束光线透过胶体，从入射光的垂直方向可以观察到胶体里出现的一条光亮的“通路”，这种现象叫（Tyndall effect）

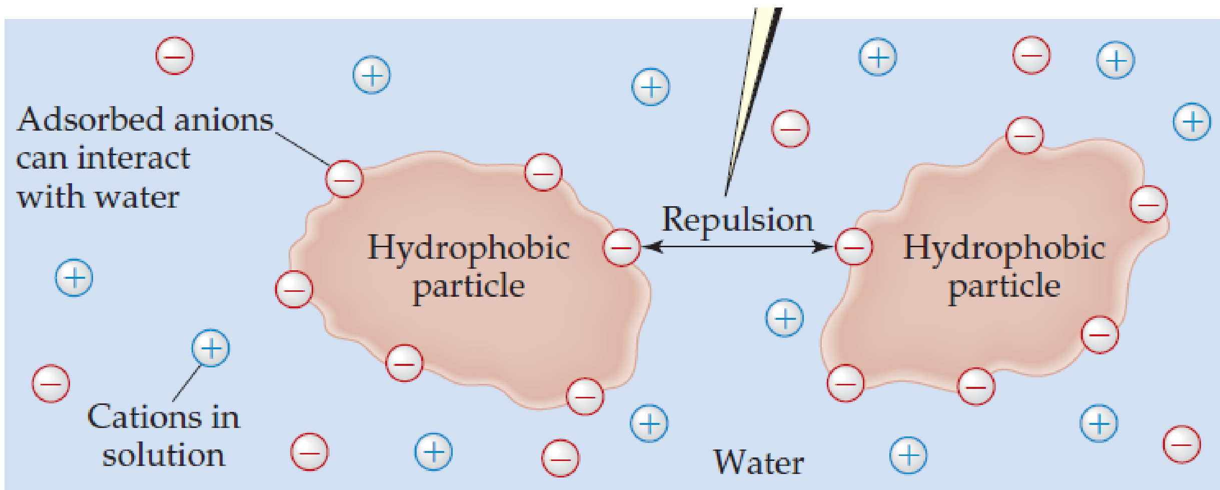


Hydrophilic and Hydrophobic Colloids

[ˌhaɪdrəˈfɪlɪk] 亲水的 [ˌhaɪdrəˈfəʊbɪk]疏水

The most important colloids are those in which the dispersing medium is water. These colloids may be hydrophilic (water loving) or hydrophobic (water fearing).





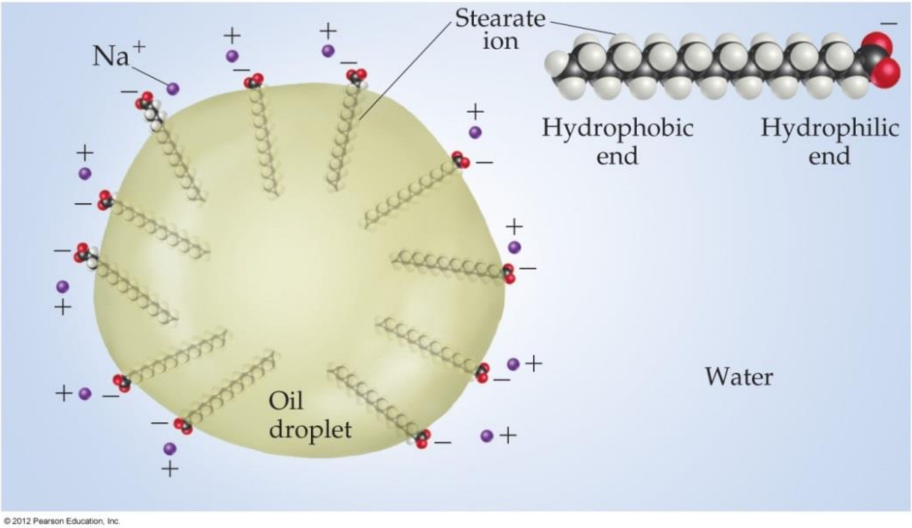
▲ **FIGURE 13.31** **Hydrophobic colloids stabilized by adsorbed anions.**

Emulsifying agents: [ɪˈmʌlsɪfaɪ]乳化剂

Some molecules have a polar, hydrophilic (*water-loving*) end and a nonpolar, hydrophobic (*water-hating*) end.

Hydrophobic groups are away from water, on the "inside" of molecules.

These molecules can help in the emulsification (乳化) of fats and oils in aqueous solutions.



Soap: $C_{17}H_{35}COONa$

Removal of Colloidal Particles

- Heating a colloidal dispersion or
- adding an electrolyte may bring about coagulation. (Coagulation: 聚沉)
- A semipermeable membrane can be used to separate ions from colloidal particles (Dialysis：渗析)

Key Equations

- $S_g = kP_g$ [13.4] Henry's law, which relates gas solubility to partial pressure
- $\text{Mass \% of component} = \dfrac{\text{mass of component in soln}}{\text{total mass of soln}} \times 100$ [13.5] Concentration in terms of mass percent
- $\text{ppm of component} = \dfrac{\text{mass of component in soln}}{\text{total mass of soln}} \times 10^6$ [13.6] Concentration in terms of parts per million (ppm)
- $\text{Mole fraction of component} = \dfrac{\text{moles of component}}{\text{total moles of all components}}$ [13.7] Concentration in terms of mole fraction
- $\text{Molarity} = \dfrac{\text{moles of solute}}{\text{liters of soln}}$ [13.8] Concentration in terms of molarity
- $\text{Molality} = \dfrac{\text{moles of solute}}{\text{kilograms of solvent}}$ [13.9] Concentration in terms of molality
- $P_{\text{solution}} = X_{\text{solvent}}P^{\circ}_{\text{solvent}}$ [13.10] Raoult's law, calculating vapor pressure of solvent above a solution
- $\Delta T_b = iK_b m$ [13.12] Calculating the boiling-point elevation of a solution
- $\Delta T_f = -iK_f m$ [13.13] Calculating the freezing-point depression of a solution
- $\Pi = i\left(\dfrac{n}{V}\right)RT = iMRT$ [13.14] Calculating the osmotic pressure of a solution

SAMPLE INTEGRATIVE EXERCISE Putting Concepts Together

A 0.100-L solution is made by dissolving 0.441 g of $CaCl_2(s)$ in water. (**a**) Calculate the osmotic pressure of this solution at 27 °C, assuming that it is completely dissociated into its component ions. (**b**) The measured osmotic pressure of this solution is 2.56 atm at 27 °C. Explain why it is less than the value calculated in (a), and calculate the van't Hoff factor, *i*, for the solute in this solution. (**c**) The enthalpy of solution for $CaCl_2$ is $\Delta H = -81.3$ kJ/mol. If the final temperature of the solution is 27 °C, what was its initial temperature? (Assume that the density of the solution is 1.00 g/mL, that its specific heat is 4.18 J/g-K, and that the solution loses no heat to its surroundings.)

The rule of "like dissolves like" refers to similarities between ________ of miscible liquids.

a. molecular weights
b. shapes
c. intermolecular attractive forces
d. densities

Solutions

Which of the following compounds is miscible with water?

a. CH_3OH
b. CH_4
c. C_6H_6
d. $CH_3CH_2OCH_2CH_3$

Solutions

Which of the compounds below is the LEAST miscible with water?

a. CH_3OH
b. CH_3CH_2OH
c. $CH_3CH_2CH_2OH$
d. $CH_3CH_2CH_2CH_2OH$

Solutions

________ Law says that the solubility of a gas in a liquid increases as the pressure of the gas increases.

a. Boyle's
b. Charles's
c. Henry's
d. Raoult's

_________ Law says that the vapor pressure of a solution is proportional to the mole fraction of the solvent.

a. Boyle’s
b. Charles’s
c. Henry’s
d. Raoult’s

Solutions

The molality of a solution is defined as the amount of solute (in moles) divided by the

a. volume of the solution (in liters).
b. mass of the solvent (in kilograms).
c. mass of the solution (in kilograms).
d. total number of moles.

Solutions

In general, as the temperature of a solution increases, the solubility of a gaseous solute

a. increases.

b. decreases.

c. remains unchanged.

d. varies from gas to gas.

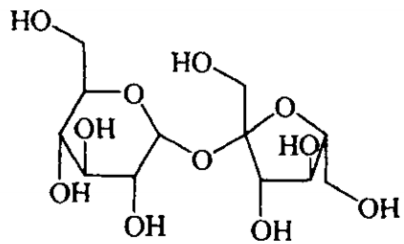
What is the freezing point of a 2.00 m aqueous solution of sucrose? The value of K_f for water is 1.86 degrees C/molal.

a. –3.72 degrees C
b. –1.86 degrees C
c. +1.86 degrees C
d. +3.72 degrees C

蔗　糖

Zhetang

Sucrose



$C_{12}H_{22}O_{11}$　342.30

[57-30-1]

本品为β-D-呋喃果糖基-α-D-吡喃葡萄糖苷。

What is the vapor pressure of a solution containing 0.500 mol of glucose and 5.000 mol of water at 29 degrees C?

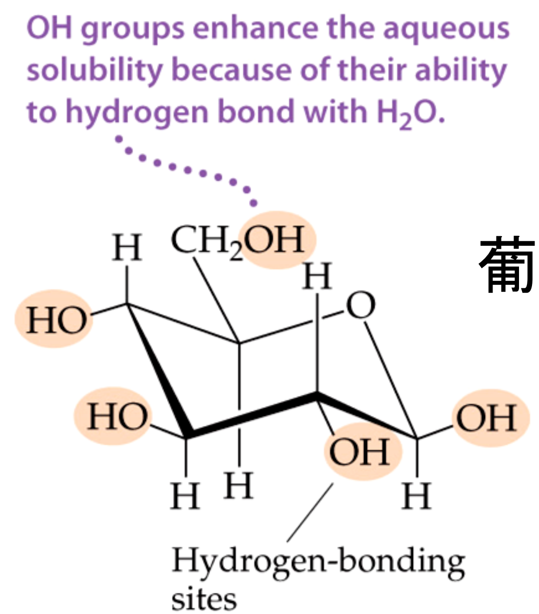
a. 30.0 Torr

b. 27.3 Torr

c. 25.0 Torr

d. 2.7 Torr

葡萄糖



Glucose, $C_6H_{12}O_6$, has five OH groups and is highly soluble in water

A 0.100 molal solution of which compound below will have the lowest freezing point?

a. NaCl
b. $CaCl_2$
c. KI
d. $LiNO_3$

Solutions

Two solutions that are isotonic have the same

a. density.
b. volume.
c. vapor pressure.
$\boxed{\text{d.}}$ osmotic pressure.

Solutions

At 298 K, 25.00 mL of solution containing 27.55 mg of protein has an osmotic pressure of 3.22 Torr. The molecular weight of the protein is

a. 2,340,000.
b. 159,000.
c. 6360.
d. 254.

R 的值	单位
8.3144621(75)	$J \cdot K^{-1} \cdot mol^{-1}$
0.08205746(14)	$L \cdot atm \cdot K^{-1} \cdot mol^{-1}$
$8.20574587 \times 10^{-5}$	$m^3 \cdot atm \cdot K^{-1} \cdot mol^{-1}$
8.3144621(75)	$cm^3 \cdot MPa \cdot K^{-1} \cdot mol^{-1}$
8.3144621(75)	$L \cdot kPa \cdot K^{-1} \cdot mol^{-1}$
8.3144621(75)	$m^3 \cdot Pa \cdot K^{-1} \cdot mol^{-1}$
62.36367(11)	$L \cdot mmHg \cdot K^{-1} \cdot mol^{-1}$
62.36367(11)	$L \cdot Torr \cdot K^{-1} \cdot mol^{-1}$
1.9858775(34)	$cal \cdot K^{-1} \cdot mol^{-1}$
8.63×10^{-5}	$eV \cdot K^{-1} \cdot atom^{-1}$

What is the molecular weight of adrenaline if 0.64 grams of adrenaline dissolved in 36.0 g of CCl_4 raises the boiling point by 0.49 degrees Celsius? (K_b = 5.02 $^oC/m$)

$\boxed{\text{a.}}$ 180 g/mole   b. 360 g/mole

c. 720 g/mole   d. 1800 g/mole

Which substance below is NOT a colloid?

a. Butter
b. Smoke
c. Whipped cream
d. $\boxed{\text{d.}}$ Salt water

Solutions

Light scattering by colloidally dispersed particles is an example of the _________ effect.

a. Tyndall
b. Raoult
c. Hall
d. Meissner

Solutions